

K-FIELD LUCAS SUPERGROUP

EXACT DETERMINISTIC RECIPROCAL FIELD SPECTRUM

Metric-weighted cellular boundary operators, complete k-form Hodge spectra, circulation closure, laminar null-mode resolution, and the 864-direction primitive reciprocal shell

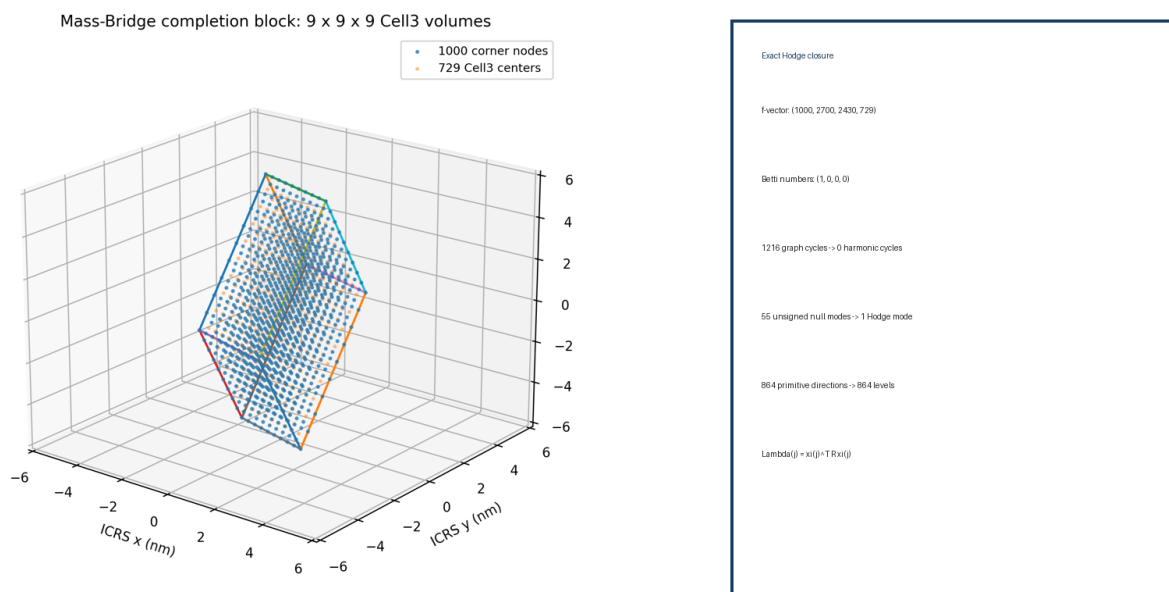
Principal Investigator: P. Dwayne Esterline

BS, Huntington University (1999) | MBA, Indiana Wesleyan University (2008)

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Title figure. The dimensionally correct 9B Lucas Supergroup contains 729 Cell3 volumes. The exact chain complex has f-vector (1000, 2700, 2430, 729), while the primitive reciprocal outer shell contains 864 antipodal directions and 864 distinct metric levels.

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Abstract

This paper constructs the complete metric-weighted cellular Hodge program of the Lucas Supergroup, defined as the 9B space-filling block containing 729 Cell3 units. The cellular complex contains 1,000 vertices, 2,700 oriented edges, 2,430 oriented faces, and 729 oriented volumes. Explicit integer boundary matrices satisfy the chain identities $B_1 B_2 = 0$ and $B_2 B_3 = 0$. Their ranks are 999, 1,701, and 729, giving Betti numbers (1,0,0,0). The filled Supergroup is therefore connected and contractible: its only harmonic cochain is the constant scalar mode.

The locked oblique Cell3 metric determines complementary Hodge stars $H_0=V$, $H_1=V/R$, $H_2=M/V$, and $H_3=1/V$. Separation across the three primitive directions reduces every positive k -form eigenvalue to one quadratic symbol, $\Lambda(j)=\xi(j)^T R \xi(j)$, where $\xi_j=2 \sin(\pi j/20)$. The degree multiplicity is $\text{binomial}(r(j),k)$, producing exact exterior-algebra multiplets $1+1$, $1+2+1$, and $1+3+3+1$. This reproduces all 6,859 cellular degrees of freedom and gives the complete scalar, vector, bivector, and volume spectra.

The 1,216 cycles of the Cell3-center graph are exactly the boundaries of dual faces; none survives as a topologically protected harmonic circulation. The 55 zero modes of the unsigned $19 \times 19 \times 19$ barycentric adjacency graph distribute across scalar, vector, and bivector sectors, but all are lifted when correct oriented incidence is restored. The oriented Hodge complex instead retains one constant scalar harmonic.

The 864 primitive antipodal reciprocal directions of the max-norm 9 shell generate 864 distinct values $q(s)=s^T R s$. Periodic 9B boundary conditions fix $\kappa_s^2=(2\pi/9)^2 q(s)$, producing a finite, anisotropic, coefficient-free spectrum of relative field levels. Geometry therefore determines a deterministic field spectrum once one universal dispersion law is declared. A unique particle assignment additionally requires a predeclared form degree, representation, polarization rule, and address-to-state grammar.

Central result: H^0 is one-dimensional, $H^1=H^2=H^3=0$, and every positive k -form level is generated by a single locked reciprocal quadratic form.

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1. Definition, scope, and mathematical contract

The Lucas Supergroup is the 9B block formed by placing nine complete Cell3 units along each primitive Cell3 direction. It therefore contains $9 \times 9 \times 9 = 729$ Cell3 volumes and 27 Lucas Groups. The present work treats the Supergroup as a finite oriented cubical complex embedded by the locked affine Cell3 basis. Topological statements follow from incidence. Metric statements follow from the locked direct and reciprocal metrics. Spectral statements follow from the resulting metric adjoints and Hodge Laplacians.

$$\text{Lucas Supergroup} = 9B; 9^3 = 729 \text{ Cell3 volumes}$$

Four spaces are distinguished throughout. C0 contains scalar values on vertices. C1 contains oriented edge cochains and is the vector or circulation sector. C2 contains oriented face cochains and is the bivector or flux sector. C3 contains oriented volume cochains and is the pseudoscalar volume sector. The boundary maps lower cell dimension, while coboundaries raise cochain degree.

The phrase deterministic reciprocal field spectrum has a precise meaning here. The geometry fixes all relative reciprocal levels without assigning a separate coefficient to any direction or object. A universal field equation may supply one propagation speed, one stiffness, or one mass term. Such constants belong to the field theory as a whole, not to individual spectral levels.

2. Locked Cell3 geometry

The Cell3 basis is the local geometric generator. Its columns are the primitive vectors a , b , and c in ICRS Cartesian coordinates, expressed in nanometres. The direct metric is $M=B^T B$. The reciprocal metric $R=M^{-1}$ determines quadratic reciprocal magnitudes. The Cell3 volume is the absolute determinant of B .

Locked Cell3: 27-node midpoint grammar and authoritative edge basis

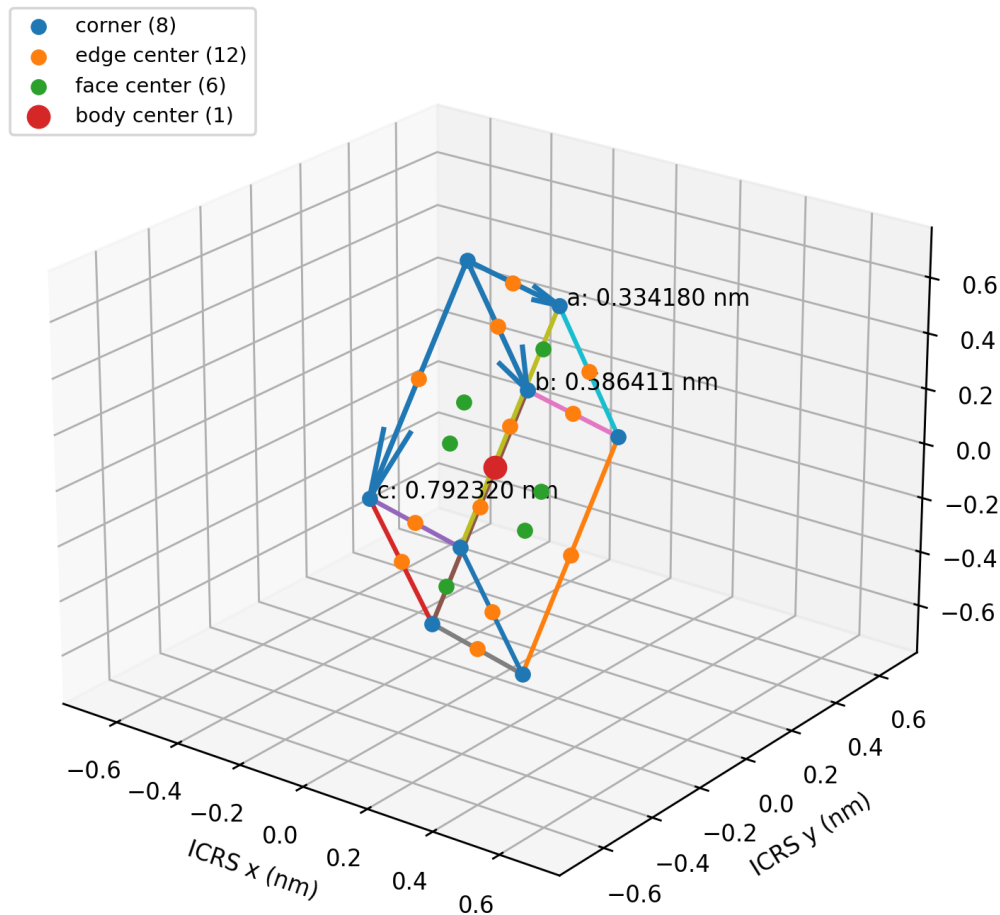


Figure 2.1. Dimensionally correct Cell3 with all 27 half-step nodes. Each address L in $\{-1,0,1\}^3$ is plotted at $p_L=B L/2$. Node polarity alternates with the sum of address components.

2.1 Ordered basis and direct metric

	x	y	z
x	0.327501242056	0.035330775844	-0.010129317627
y	-0.054897007884	0.20731985044	-0.389787702366
z	-0.037487502895	-0.547401085389	-0.689734408367

Table 2.1. Locked basis B in nanometres; rows are ICRS x, y, z and columns are primitive directions a, b, c.

	a	b	c
a	0.111676057896	0.020710333281	0.043937235096
b	0.020710333281	0.343877732393	0.296392758962
c	0.043937235096	0.296392758962	0.627770610077

Table 2.2. Direct Gram metric M in square nanometres.

2.2 Reciprocal metric and primitive measures

	a*	b*	c*
a*	9.208025529458	0.001534802428	-0.6451880357
b*	0.001534802428	4.903398312368	-2.315175584416
c*	-0.6451880357	-2.315175584416	2.731171274712

Table 2.3. Reciprocal metric R in inverse square nanometres.

Quantity	Value
Cell3 volume	0.117914891911745 nm ³
Edge a	0.334179679059111 nm
Edge b	0.586410890411141 nm
Edge c	0.792319765042685 nm
Angle ab	83.933483012313 deg
Angle ac	80.448130018338 deg
Angle bc	50.363231813756 deg

Table 2.4. Primitive Cell3 measures.

Complete Cell3 address atlas: all 27 nodes, lambda values, and polarities

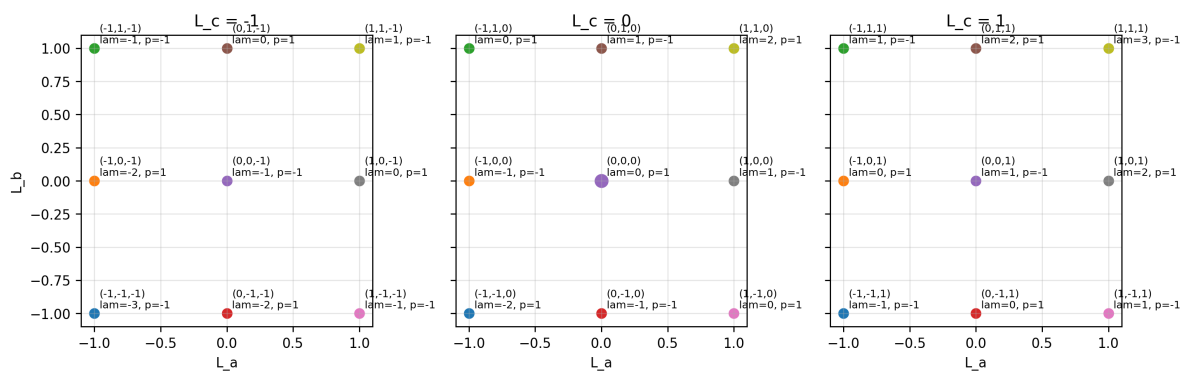


Figure 2.2. Complete address atlas for all 27 Cell3 half-step nodes. The three panels fix the third primitive address component.

Authoritative Cell3 boundary-plane atlas: six faces, nine nodes per face

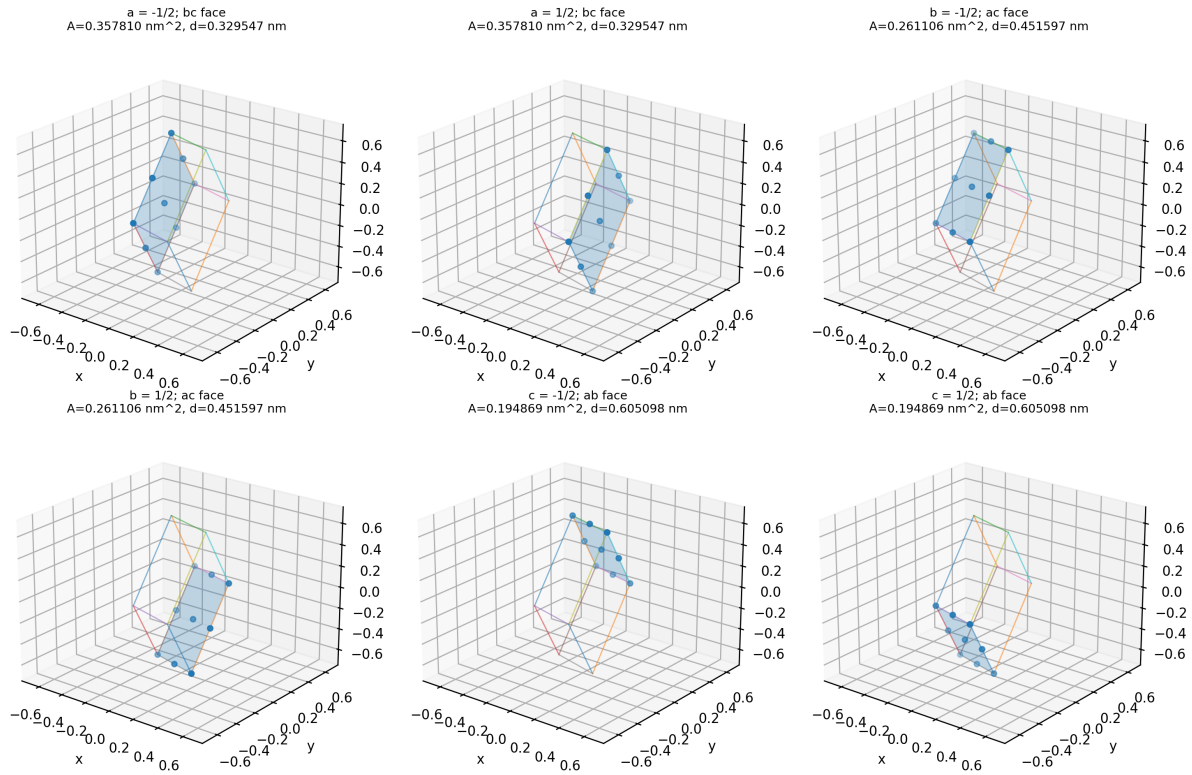


Figure 2.3. Dimensionally correct atlas of the six Cell3 boundary planes. Each opposite pair is normal to one reciprocal primitive direction and contains the nine appropriate half-step nodes.

3. Lucas Supergroup cellular complex

Mass-Bridge completion block: 9 x 9 x 9 Cell3 volumes

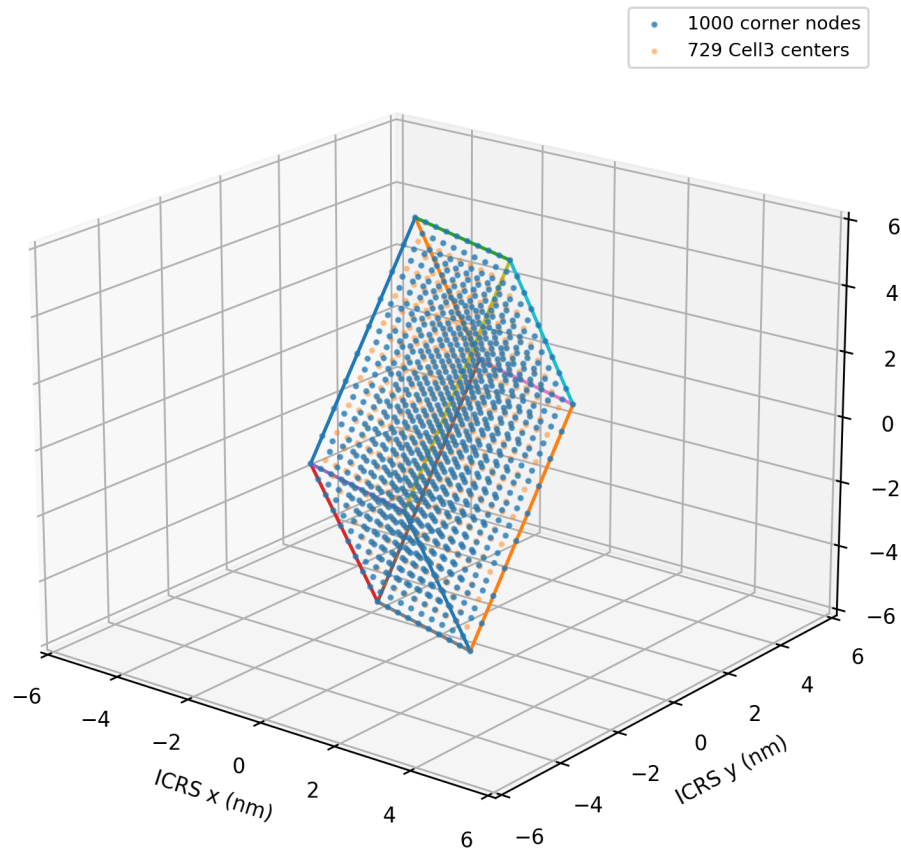


Figure 3.1. Dimensionally correct 9B Lucas Supergroup. The block contains 729 Cell3 volumes, 1,000 merged corner vertices, and 27 embedded Lucas Groups.

The f-vector counts cells by dimension. With nine intervals per primitive axis, there are ten boundary-coordinate positions and nine cell-center positions. A k-cell selects k center coordinates and 3-k boundary coordinates. Therefore $f_k = \text{binomial}(3,k) 9^k 10^{(3-k)}$.

Degree k	Object	Formula	Count
0	vertices	10^3	1,000
1	oriented edges	$3 \times 9 \times 10^2$	2,700
2	oriented faces	$3 \times 9^2 \times 10$	2,430
3	Cell3 volumes	9^3	729
sum	all barycentric states	$(10+9)^3$	6,859
$F(t) = (10 + 9 t)^3 = 1000 + 2700 t + 2430 t^2 + 729 t^3$			

The alternating evaluation $F(-1)=1$ is the Euler characteristic of the filled three-dimensional block.

g=2 merged half-step lattice: 6,859 nodes on 55 lambda sheets

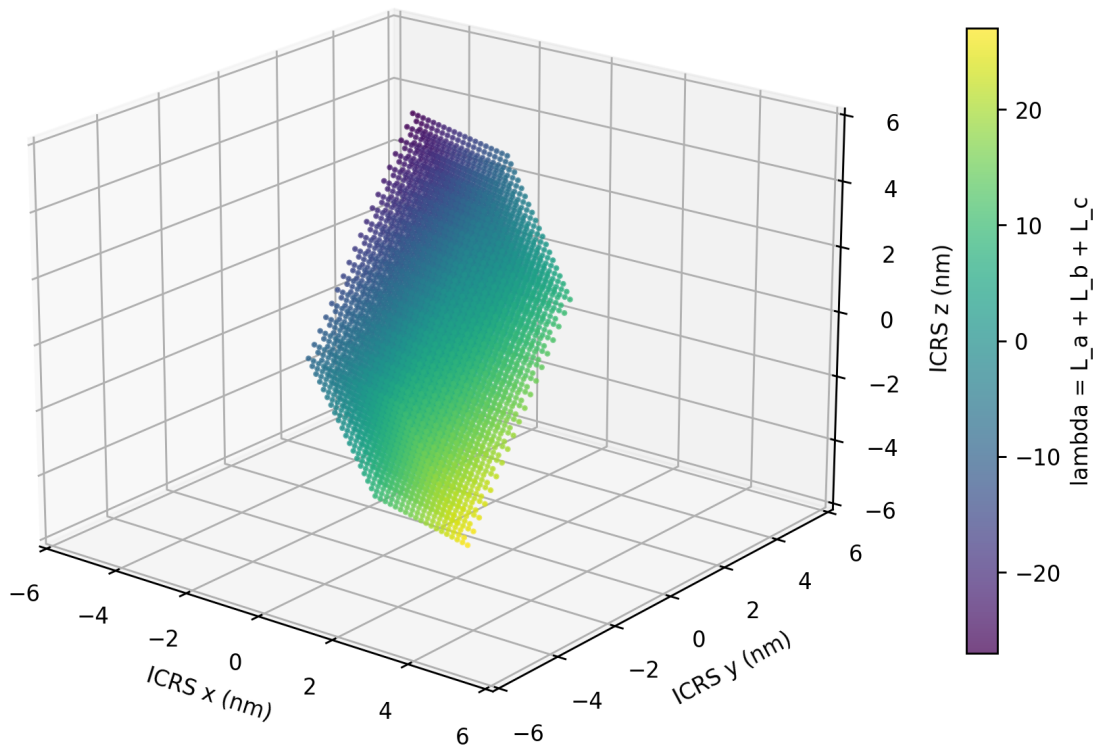
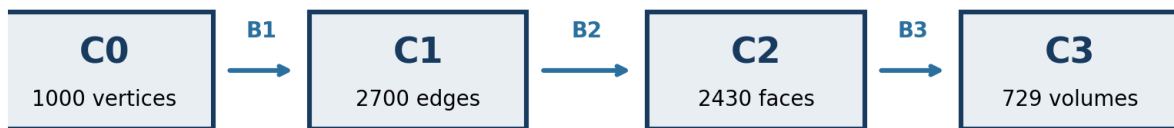


Figure 3.2. The complete 19 x 19 x 19 barycentric lattice. Each of the 6,859 points represents exactly one vertex, edge, face, or volume barycenter.

4. Exact cellular boundary operators



$$B1 \ B2 = 0 \quad \text{and} \quad B2 \ B3 = 0$$

$$\text{rank}(B1, B2, B3) = (999, 1701, 729) \quad \text{Betti} = (1, 0, 0, 0)$$

Figure 4.1. Exact oriented cellular chain complex. The boundary maps satisfy boundary-of-boundary equals zero.

4.1 Orientation conventions

Every primitive edge is oriented in the positive a, b, or c direction. Faces use the ordered orientations a wedge b, a wedge c, and b wedge c. Volumes use a wedge b wedge c. With these conventions, each edge boundary has one positive and one negative endpoint, each face boundary has four signed edges, and each volume boundary has six signed faces.

$$\text{partial}_1 = B1, \text{partial}_2 = B2, \text{partial}_3 = B3$$

Operator	Shape	Nonzeros	Rank	Meaning
B1	1000 x 2700	5,400	999	edge endpoints
B2	2700 x 2430	9,720	1,701	face-edge incidence
B3	2430 x 729	4,374	729	volume-face incidence

4.2 Chain closure and homology

$$B1 \ B2 = 0; \ B2 \ B3 = 0$$

The rank-nullity calculations give Betti numbers $\beta=(1,0,0,0)$. The single β_0 mode records connectedness. Vanishing β_1 , β_2 , and β_3 show that the filled Supergroup contains no topological tunnel, enclosed void, or free three-dimensional volume cycle.

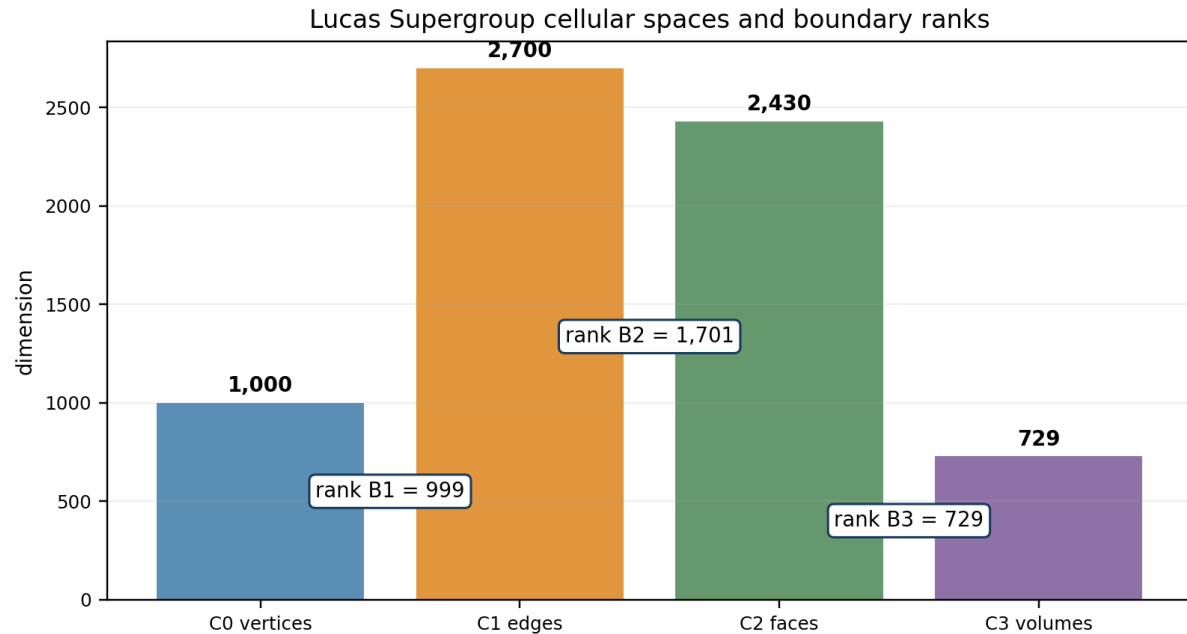


Figure 4.2. Cellular-space dimensions and the ranks of the three boundary operators.

5. Metric Hodge construction

Incidence fixes topology, but positive spectral values require a metric. The affine map $x=B \ u$ induces the direct metric M and reciprocal metric R . In the complementary ordered form bases $1; (du^a, du^b, du^c); (du^b \wedge du^c, du^c \wedge du^a, du^a \wedge du^b);$ and $du^a \wedge du^b \wedge du^c$, the constant modal Hodge stars are $H0=V$, $H1=V \ R$, $H2=M/V$, and $H3=1/V$.

$$H0 = V; \ H1 = V \ R; \ H2 = M / V; \ H3 = 1 / V$$

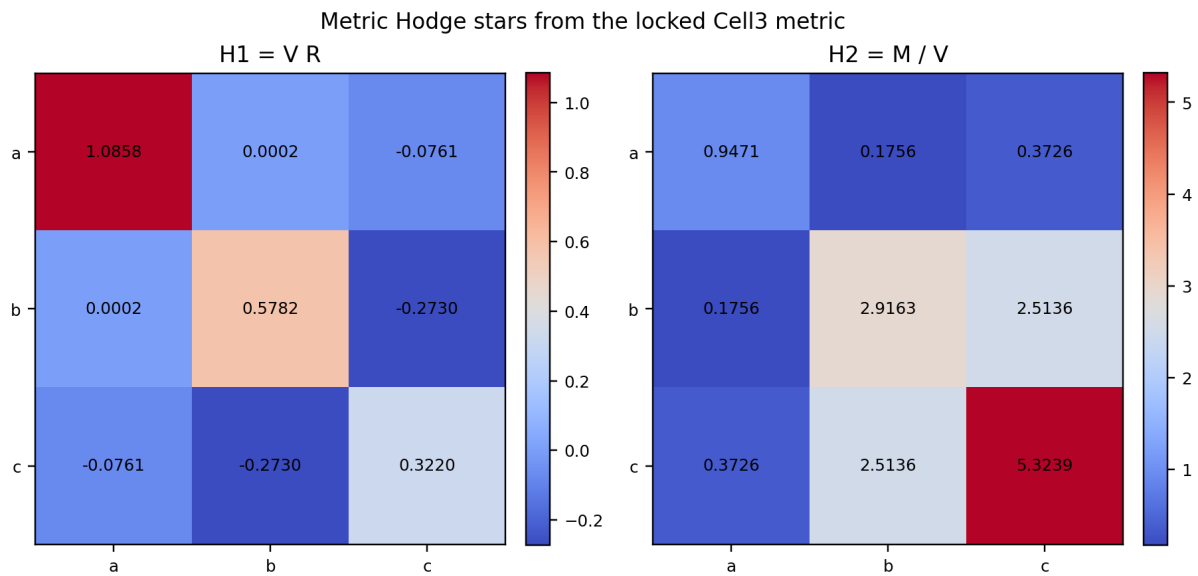


Figure 5.1. Metric Hodge-star matrices. H1 weights one-forms by the reciprocal metric; H2 weights complementary two-forms by the direct metric.

5.1 Numerical Hodge stars

$H_0 = 0.117914891911745$ and $H_3 = 8.480693013300360$.

	a	b	c
a	1.085763335027	0.000180976062	-0.076077277492
b	0.000180976062	0.578183682003	-0.272993678793
c	-0.076077277492	-0.272993678793	0.32204576565

Table 5.1. One-form Hodge star $H_1=V R$.

	bc	ca	ab
bc	0.947090363952	0.175637978756	0.372618202704
ca	0.175637978756	2.916321482533	2.513616000118
ab	0.372618202704	2.513616000118	5.323929826838

Table 5.2. Two-form Hodge star $H_2=M/V$ in the complementary basis.

$$H_1 H_2 = I_3; H_0 H_3 = 1$$

5.2 Metric adjoints and Hodge Laplacians

$$d_k = B_{(k+1)}^T$$

$$\delta_{(k+1)} = H_k^{(-1)} d_k^T H_{(k+1)}$$

$$\Delta_k = \delta_{(k+1)} d_k + d_{(k-1)} \delta_k$$

The operators Δ_k are self-adjoint in the H_k inner products and nonnegative. Their kernels are the harmonic cochain spaces, which are isomorphic to cellular cohomology.

6. Separable modal diagonalization

The nine-cell interval has ten vertex positions. Its incidence singular symbols are $x_{i,j}=2 \sin(\pi j/20)$, $j=0,\dots,9$. Three-dimensional modes are tensor products of one-dimensional modes. For the index triple $j=(j_a,j_b,j_c)$, define $x_i(j)=(x_{i,j_a},x_{i,j_b},x_{i,j_c})$.

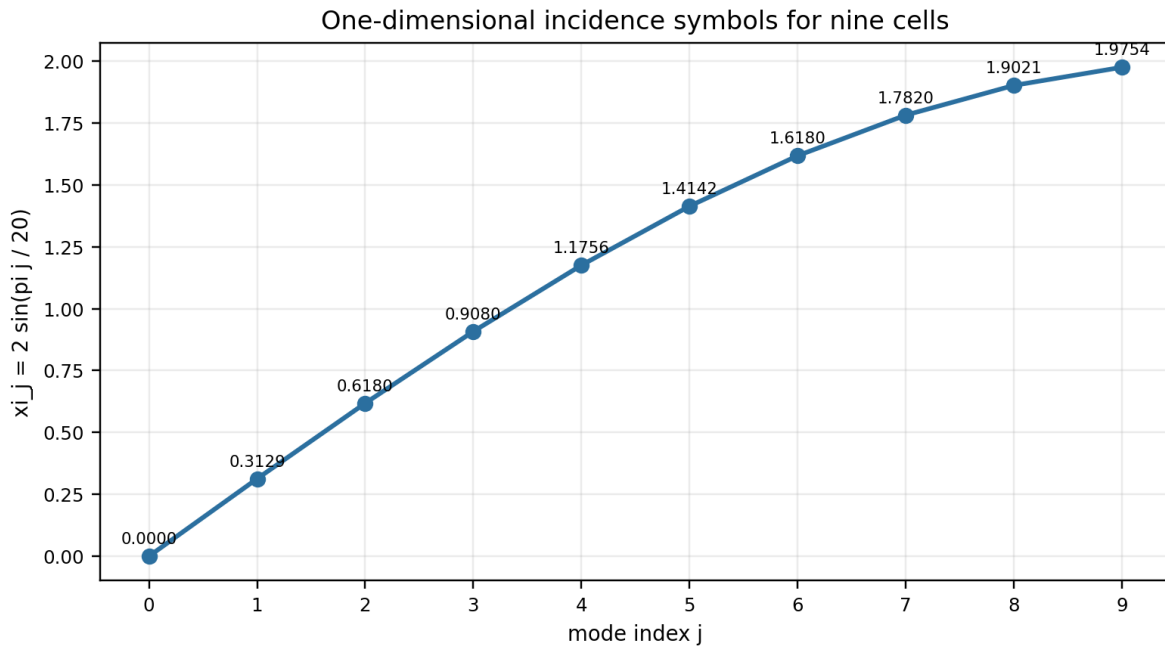


Figure 6.1. The ten exact one-dimensional incidence symbols for a nine-cell interval.

j	ξ_j
0	0.000000000000000
1	0.312868930080462
2	0.618033988749895
3	0.907980999479093
4	1.175570504584946
5	1.414213562373095
6	1.618033988749895
7	1.782013048376736
8	1.902113032590307
9	1.975376681190276

6.1 Universal metric symbol

$$\text{Lambda}(j) = \xi(j)^T R \xi(j)$$

Every positive scalar, vector, bivector, and volume eigenvalue is generated by this same reciprocal quadratic form. The form degree changes only the multiplicity, not the eigenvalue.

6.2 Exterior-algebra multiplicities

Let $r(j)$ be the number of nonzero components of j . The eigenvalue appears in degree k with multiplicity $\text{binomial}(r, k)$. This is the dimension of the k -th exterior power of the active directional subspace.

$$m_k(j) = \text{binomial}(r(j), k)$$

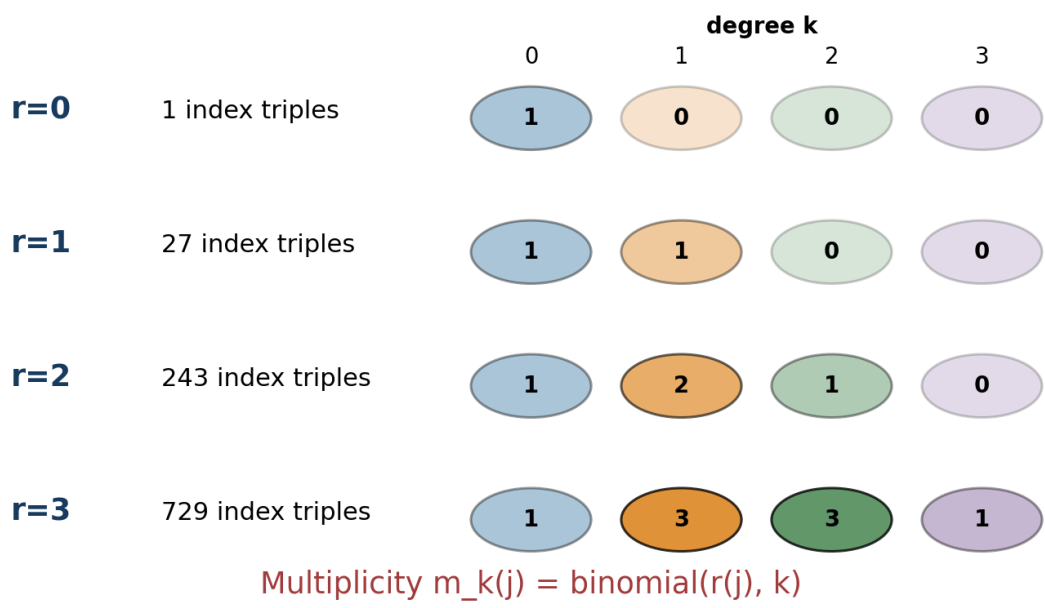


Figure 6.2. Pascal-row multiplicities for modal rank r . Fully three-directional modes form the 1+3+3+1 multiplet.

Active rank r	Index triples	$k=0$	$k=1$	$k=2$	$k=3$
0	1	1	0	0	0
1	27	1	1	0	0
2	243	1	2	1	0
3	729	1	3	3	1

Summing over all index triples gives dimensions (1000,2700,2430,729), exactly reproducing the cellular f-vector.

7. Complete primal k-form Hodge spectra

Complete metric-weighted primal Hodge spectra

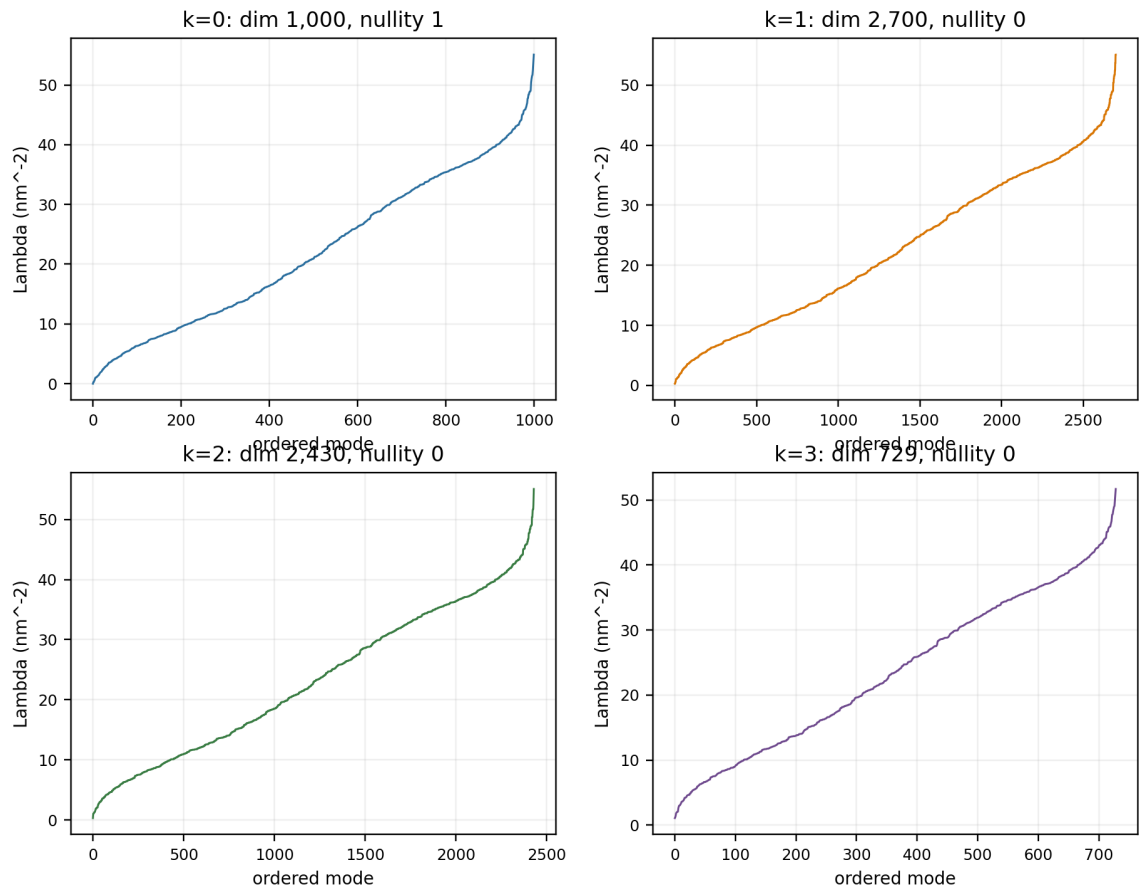


Figure 7.1. Complete ordered spectra in degrees k=0,1,2,3. All positive levels are generated by the same locked quadratic symbol.

k	Sector	Dimension	Nullity	Distinct levels	Lowest positive	Maximum	Trace
0	scalar	1,000	1	1,000	0.267346074	55.076348827	22,207.395498
1	vector/circulation	2,700	0	999	0.267346074	55.076348827	61,369.779825
2	bivector/flux	2,430	0	972	0.29407383	55.076348827	56,420.539866
3	volume pseudoscalar	729	0	729	1.069408999	51.684484228	17,258.155540

Table 7.1. Complete spectral summaries; eigenvalues are in inverse square nanometres.

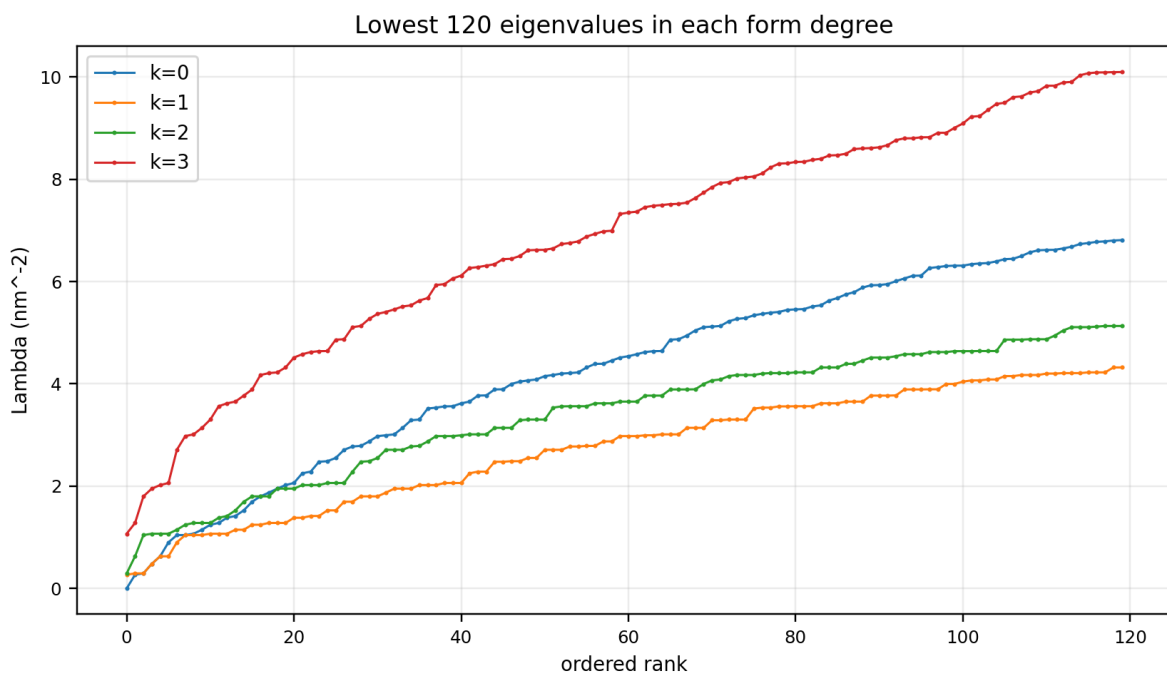


Figure 7.2. Low-energy spectral comparison. The first volume-form mode requires all three primitive directions to be active.

Index j	xi vector	r	Lambda (nm ⁻²)	Degrees present
(0, 0, 0)	(0.000000, 0.000000, 0.000000)	0	0.000000000000	0
(0, 0, 1)	(0.000000, 0.000000, 0.312869)	1	0.267346073558	0, 1
(0, 1, 1)	(0.000000, 0.312869, 0.312869)	2	0.294073830399	0, 1, 2
(0, 1, 0)	(0.000000, 0.312869, 0.000000)	1	0.479978790800	0, 1
(0, 1, 2)	(0.000000, 0.312869, 0.618034)	2	0.627851865434	0, 1, 2
(1, 0, 0)	(0.312869, 0.000000, 0.000000)	1	0.901345694910	0, 1
(1, 0, 1)	(0.312869, 0.000000, 0.312869)	2	1.042380768020	0, 1, 2
(0, 0, 2)	(0.000000, 0.000000, 0.618034)	1	1.043214597842	0, 1
(1, 1, 1)	(0.312869, 0.312869, 0.312869)	3	1.069408999171	0, 1, 2, 3
(0, 2, 2)	(0.000000, 0.618034, 0.618034)	2	1.147509326142	0, 1, 2
(0, 2, 1)	(0.000000, 0.618034, 0.312869)	2	1.244936045296	0, 1, 2
(1, 1, 2)	(0.312869, 0.312869, 0.618034)	3	1.279986229793	0, 1, 2, 3
(1, 1, 0)	(0.312869, 0.312869, 0.000000)	2	1.381624960020	0, 1, 2
(0, 1, 3)	(0.000000, 0.312869, 0.907981)	2	1.416251214015	0, 1, 2
(0, 2, 3)	(0.000000, 0.618034, 0.907981)	2	1.526207347727	0, 1, 2

7.1 Degree thresholds

The scalar and vector sectors begin at 0.267346073558 nm⁻². The bivector sector begins at 0.294073830399 nm⁻² because at least two directions must be active. The volume sector begins at 1.069408999171 nm⁻² because all three directions must be active.

7.2 Spectral conditioning

The positive condition numbers are 206.011437 for k=0 and k=1, 187.287487 for k=2, and 48.329951 for k=3. The volume spectrum is therefore substantially less ill-conditioned than the scalar and vector spectra.

8. Exact Hodge decomposition and supersymmetric pairing

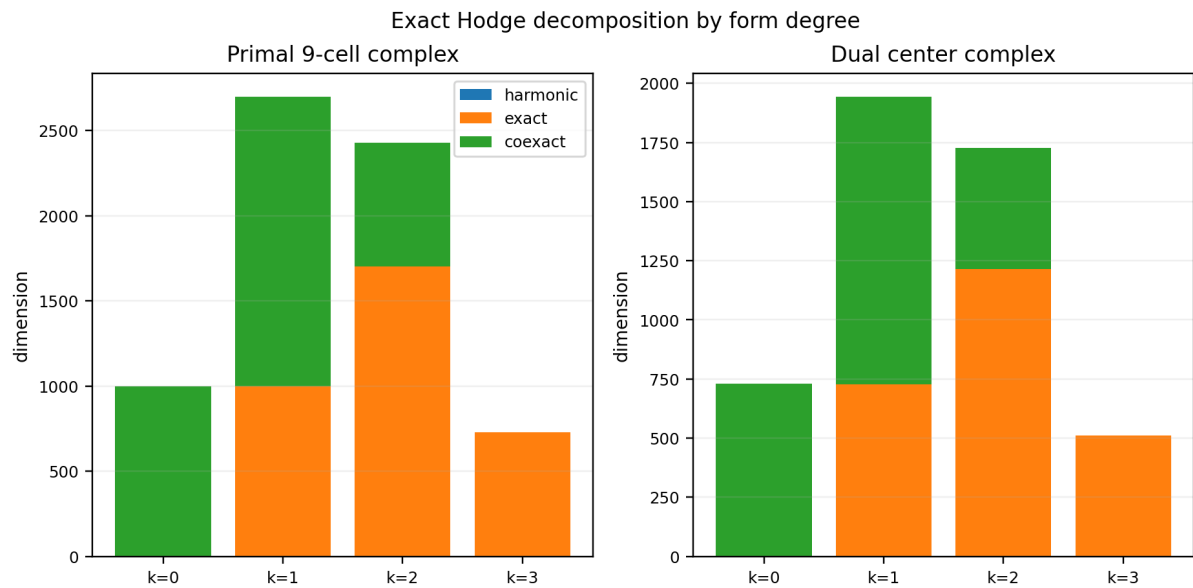


Figure 8.1. Exact, coexact, and harmonic dimensions in the primal and dual complexes.

Degree	Harmonic	Exact	Coexact	Total
0	1	0	999	1,000
1	0	999	1,701	2,700
2	0	1,701	729	2,430
3	0	729	0	729
C0 = H0 (+) im(delta1): 1000 = 1 + 999				
C1 = im(d0) (+) im(delta2): 2700 = 999 + 1701				
C2 = im(d1) (+) im(delta3): 2430 = 1701 + 729				
C3 = im(d2): 729 = 729				

Every positive modal block is paired across adjacent form degrees according to a Pascal row. The only unpaired state is the constant scalar harmonic. This finite exterior-algebra pairing is the algebraic reason the alternating heat trace remains one at all positive times.

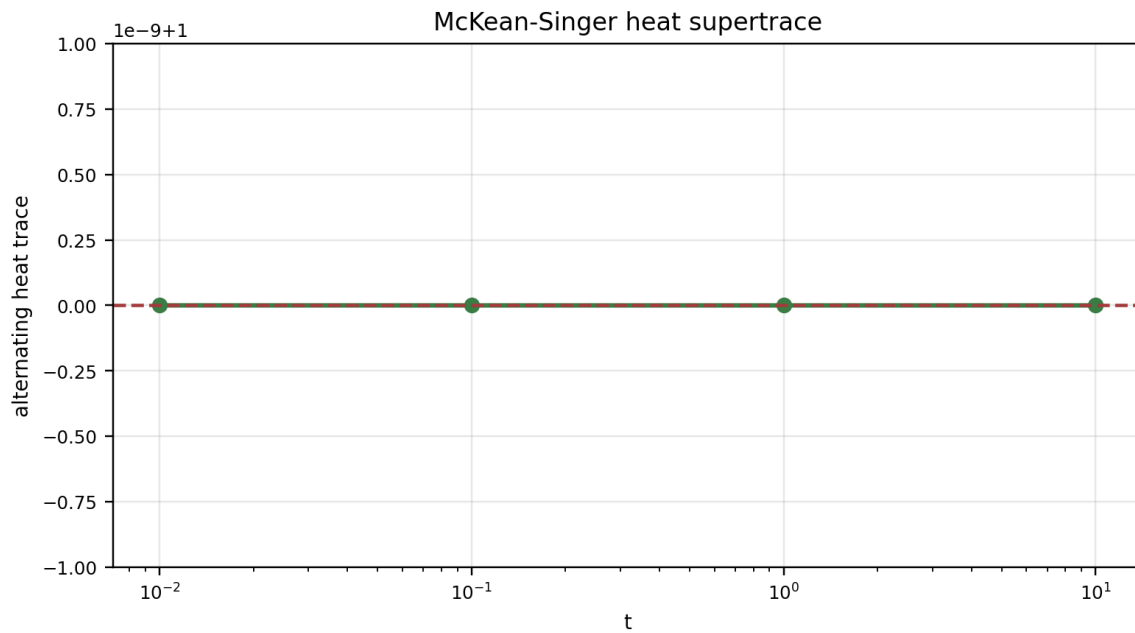


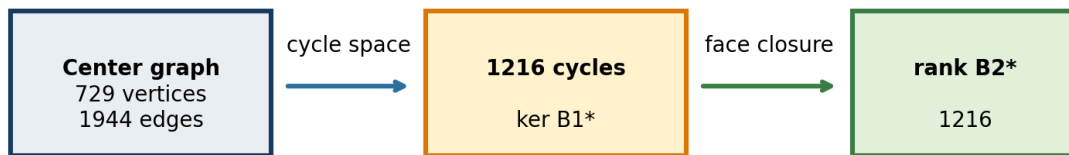
Figure 8.2. The alternating heat trace is exactly the Euler characteristic, equal to one for every tested time.

$$\sum_k (-1)^k \text{Tr}(\exp(-t \Delta_k)) = 1$$

For any positive modal block with active rank r , the alternating multiplicity is $\sum_k (-1)^k \text{binomial}(r, k) = (1-1)^r = 0$. Only the zero scalar mode remains.

9. Dual center complex and the 1,216 circulation modes

The centers of the 729 Cell3 volumes form the vertices of the dual $8 \times 8 \times 8$ cubical complex. Its f-vector is (729, 1944, 1728, 512). The center graph alone has 1,944 edges and 729 vertices, giving a graph cycle-space dimension $1,944 - 729 + 1 = 1,216$.



$$\ker B1^* = \text{im } B2^*$$

Harmonic circulation dimension = 0; all 1216 cycles are finite-energy coexact modes.

Figure 9.1. Exact face closure of the center-graph circulation space.

Quantity	Value
Dual f-vector	(729, 1944, 1728, 512)
rank B1*	728
rank B2*	1,216

Quantity	Value
rank B3*	512
graph cycle dimension	1,216
harmonic circulations after face closure	0
$\ker(B1^*) = \text{im}(B2^*)$	

No center-graph circulation survives as a harmonic one-cycle. All 1,216 circulate dynamically as curl-generated coexact modes with positive Hodge energy. They are not topologically protected because every graph cycle is a linear combination of dual face boundaries.

9.1 Dual Hodge decomposition

1944 dual 1-forms = 728 gradients + 1216 circulations + 0 harmonic

The dual scalar spectral gap is 0.329419564 nm⁻². The dual volume spectrum begins at 1.317708695 nm⁻². As in the primal complex, the dual complex is contractible and has Betti numbers (1,0,0,0).

10. Resolution of the 55 laminar adjacency-null modes

The unsigned barycentric incidence graph is the Cartesian product P19 x P19 x P19. Its adjacency nullity is 55. The zero triples are (p,10,20-p) and all coordinate permutations for p=1,...,9, together with the central triple (10,10,10). This gives 9 x 6 + 1 = 55.

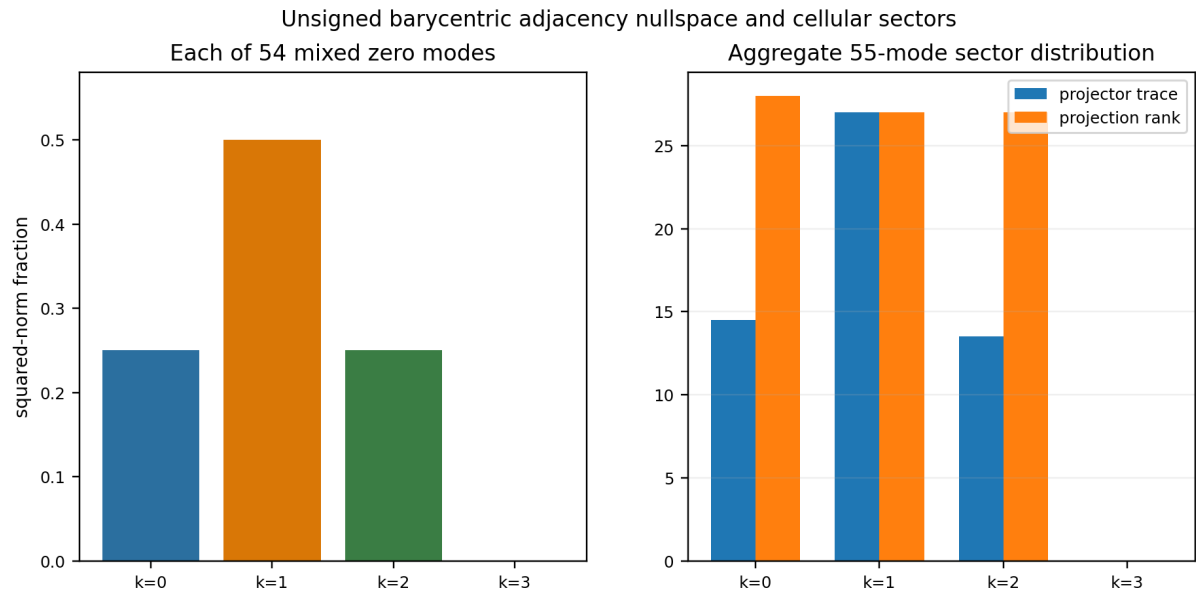


Figure 10.1. The 55 unsigned zero modes distribute over scalar, vector, and bivector sectors but have no volume-sector component.

The central mode (10,10,10) is purely scalar. Each of the other 54 modes has squared-norm fractions (1/4,1/2,1/4,0) across k=(0,1,2,3). Summed projector traces are (14.5,27,13.5,0), while the corresponding projection ranks are (28,27,27,0). Fractional projector traces measure subspace weight, not a fractional number of eigenvectors.

$$54 \times (1/4, 1/2, 1/4, 0) + (1, 0, 0, 0) = (14.5, 27, 13.5, 0)$$

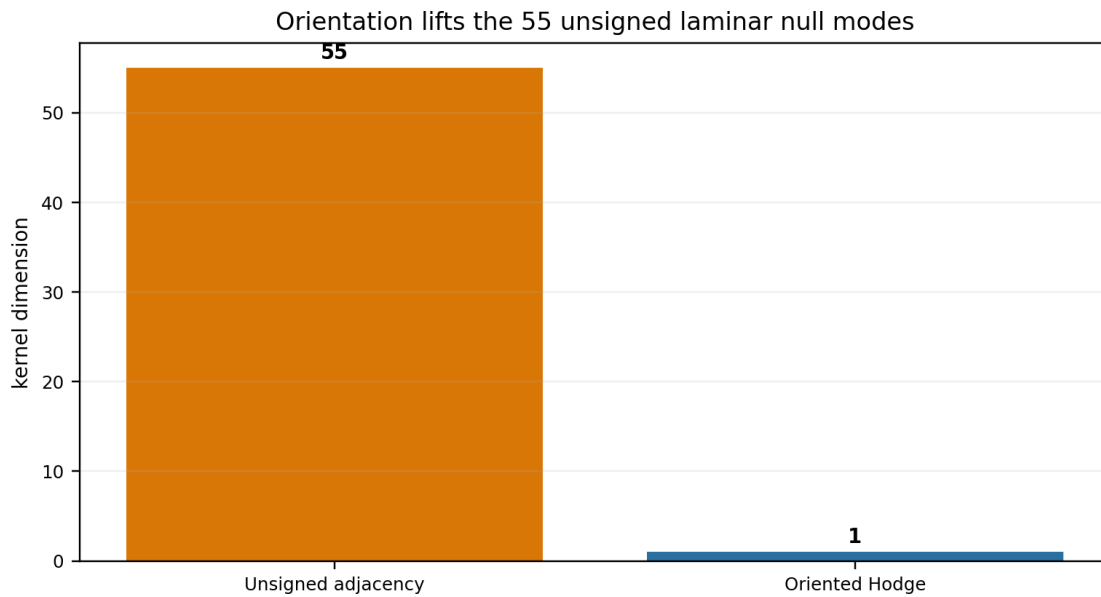


Figure 10.2. Correct oriented incidence lifts the 55 unsigned adjacency zero modes. The Hodge complex has one constant scalar kernel.

The unsigned adjacency graph and the oriented cellular Dirac/Hodge complex are different operators. When the signs required by cellular orientation are restored, all 55 unsigned modes acquire positive energy. A different state, the constant scalar cochain, becomes the sole Hodge harmonic. The nullity therefore changes from 55 to 1, but the surviving Hodge state is not one of the unsigned laminar modes.

The 55 laminar modes are real null modes of the unsigned adjacency operator, not topological harmonic forms of the oriented filled Supergroup.

11. The 864-direction primitive reciprocal shell

A primitive outer-shell address s is an integer triple with maximum absolute component 9, greatest common divisor one, and antipodal pairs identified. The shell contains exactly 864 such direction classes. For each direction, the locked reciprocal metric defines $q(s) = s^T R s$.

864 primitive antipodal directions on the max-norm 9 shell

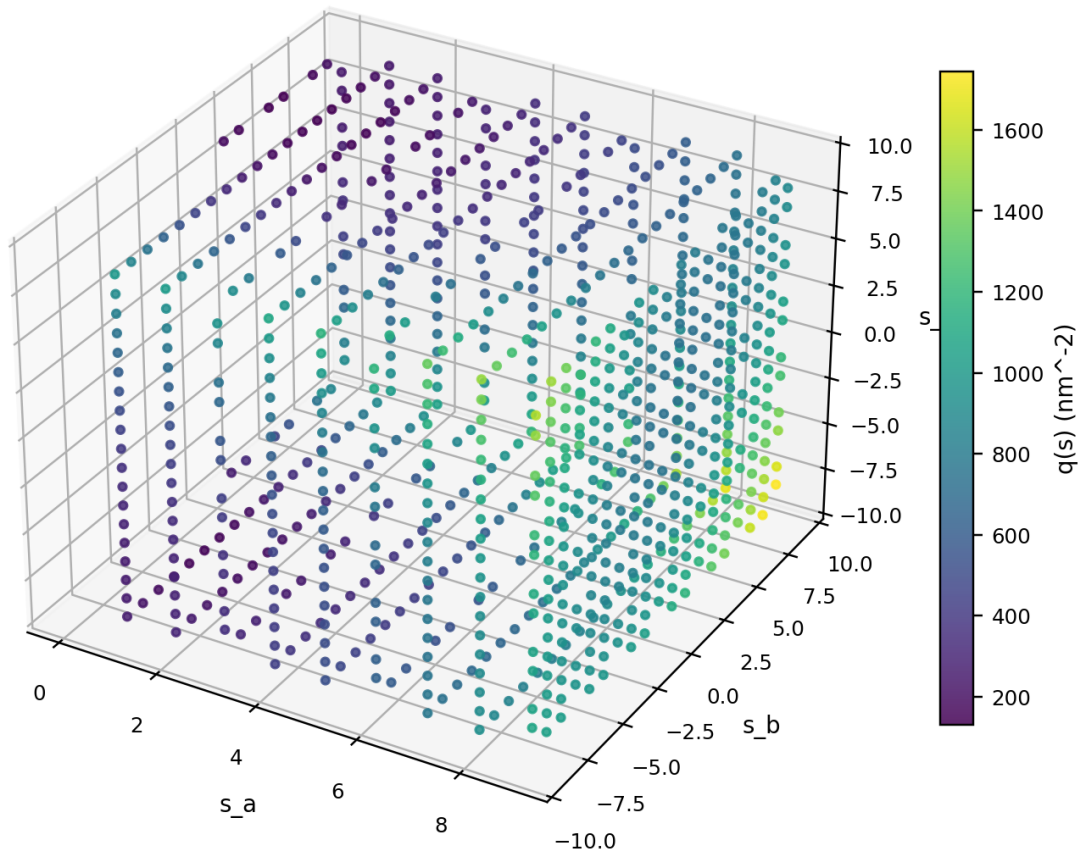


Figure 11.1. All 864 primitive antipodal directions, colored by their reciprocal quadratic value $q(s)$.

$q(s) = s^T R s$	
$\kappa_s^2 = (2 \pi / 9)^2 q(s)$	
Statistic	Value
Direction classes	864
Distinct q values to $1e-10$	864
q minimum	130.593523477839 nm ⁻²
q maximum	1744.361292059535 nm ⁻²
q mean	757.916780244175 nm ⁻²
q standard deviation	325.034198848754 nm ⁻²
Periodic κ^2 minimum	63.649699460279 nm ⁻²
Periodic κ^2 maximum	850.180537540776 nm ⁻²

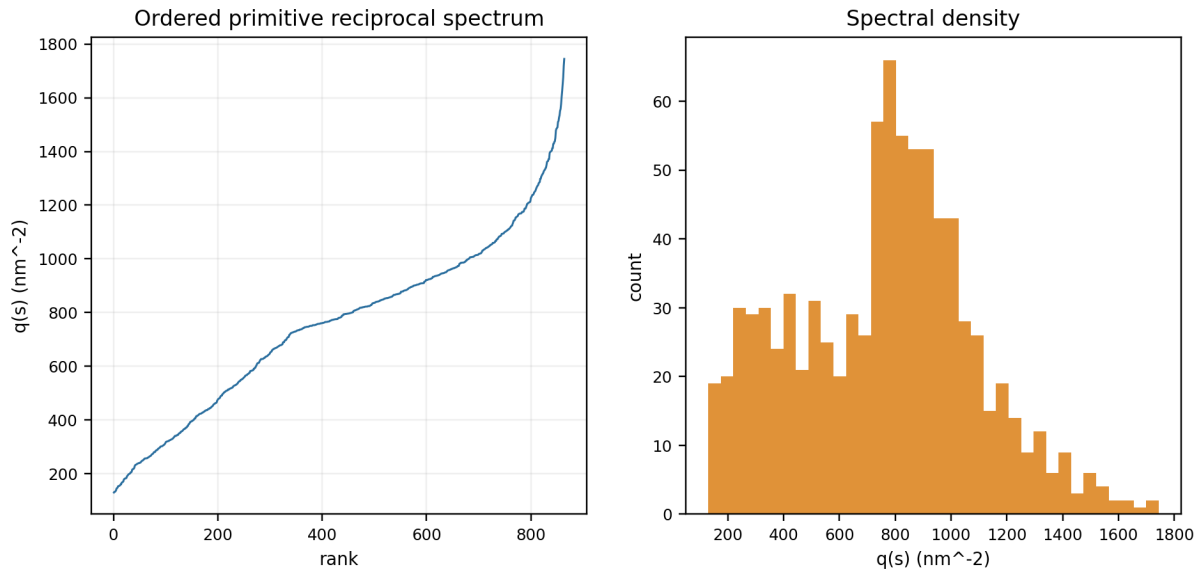


Figure 11.2. Ordered 864-level reciprocal spectrum and its finite density.

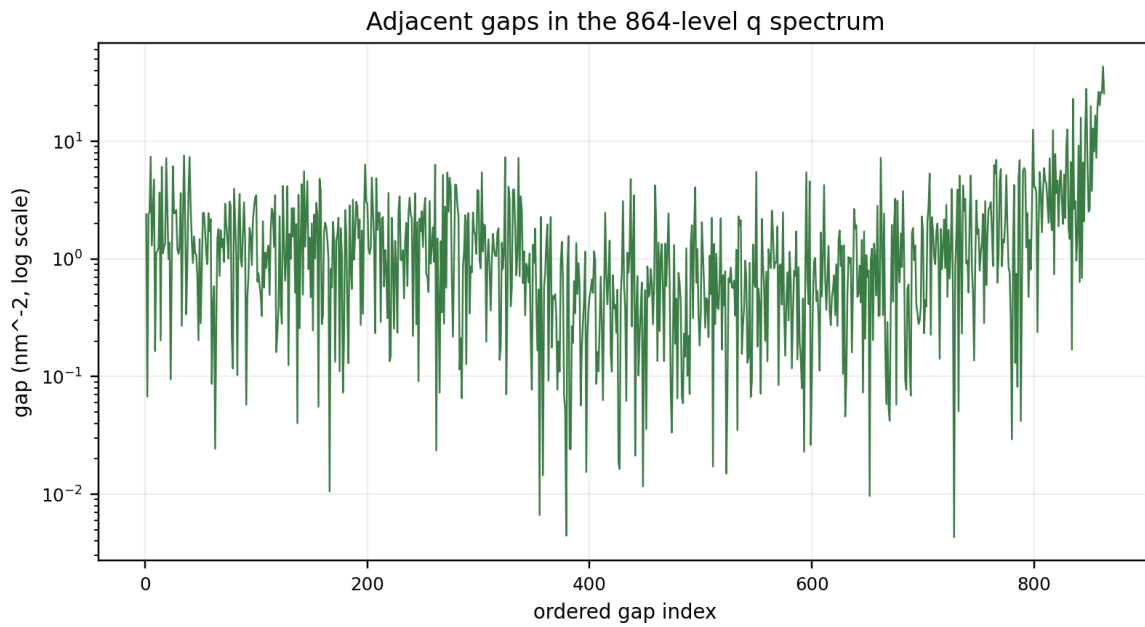


Figure 11.3. Adjacent q -level gaps. The spectrum has no exact non-antipodal degeneracy at the tested precision.

11.1 Extreme directions

The minimum occurs at $s=(1, 4, 9)$ with $q=130.593523477839 \text{ nm}^{-2}$. The maximum occurs at $s=(9, 9, -8)$ with $q=1744.361292059535 \text{ nm}^{-2}$. The closest pair of distinct levels has a relative separation of $4.060046\text{e-}06$.

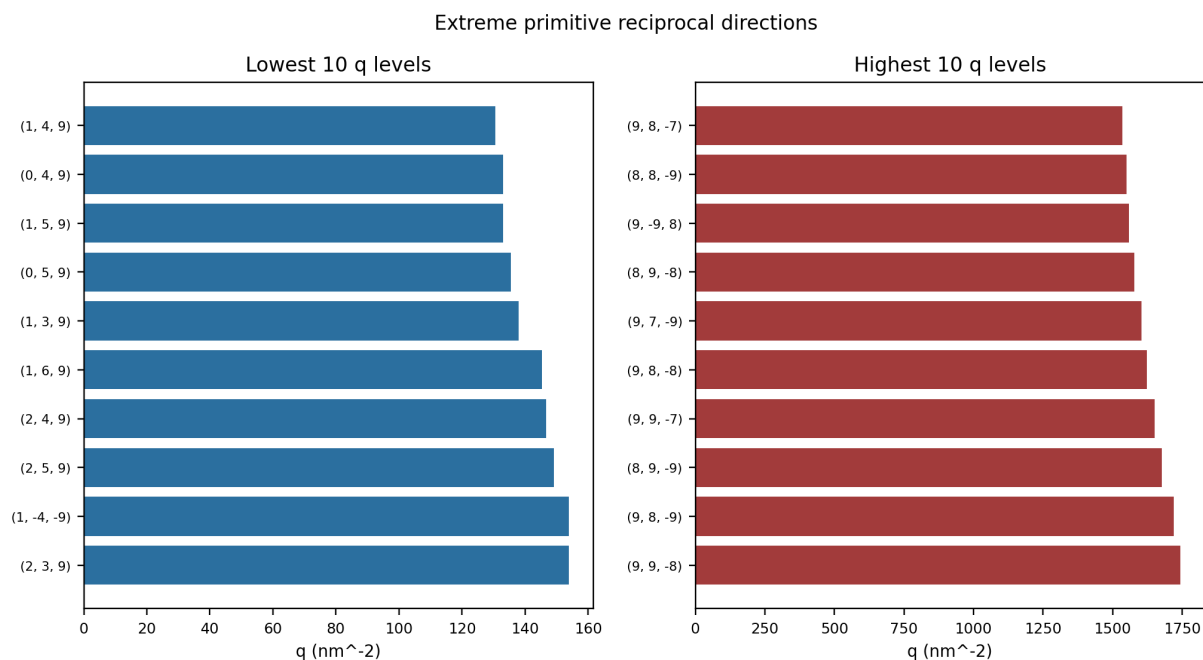
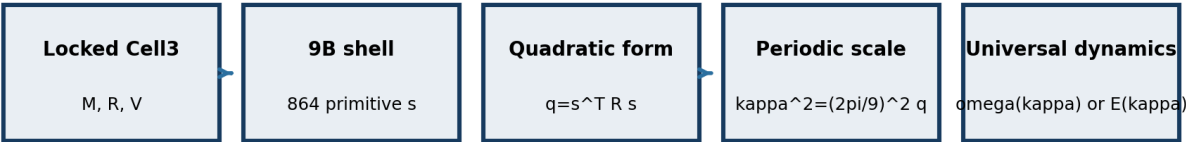


Figure 11.4. Lowest and highest primitive reciprocal levels.

11.2 Lowest 25 deterministic shell levels

Rank	Address	q (nm ⁻²)	sqrt(q/q _{min})	periodic kappa ²
1	(1, 4, 9)	130.593523477839	1.000000000000	63.649699460279
2	(0, 4, 9)	132.986604171555	1.009120731449	64.816057966297
3	(1, 5, 9)	133.054017374512	1.009376469394	64.848914343808
4	(0, 5, 9)	135.444028463372	1.018401690188	66.013776761648
5	(1, 3, 9)	137.939826205902	1.027741793138	67.230198311470
6	(1, 6, 9)	145.321307895922	1.054881872185	70.827842962057
7	(2, 4, 9)	146.616493843039	1.059572297312	71.459100884224
8	(2, 5, 9)	149.080057344568	1.068437089973	72.659811855931
9	(1, -4, -9)	153.795735924186	1.085203854193	74.958176402278
10	(2, 3, 9)	153.959726966245	1.085782271711	75.038103647238
11	(1, 2, 9)	155.092925558701	1.089770818592	75.590410897382
12	(1, -5, -9)	156.250090611146	1.093828712395	76.154399109451
13	(0, 2, 9)	157.492145462129	1.098167609137	76.759761579755
14	(1, -3, -9)	161.148177861961	1.110840946571	78.541667429824
15	(2, 6, 9)	161.350417470834	1.111537776037	78.640236562358
16	(1, 7, 9)	167.395395042067	1.132168150890	81.586485315026
17	(1, -6, -9)	168.511241922843	1.135935363167	82.130335551344
18	(0, 7, 9)	169.779266921214	1.140201235500	82.748355556510
19	(2, 2, 9)	171.109756714187	1.144660159271	83.396820144971
20	(1, -2, -9)	178.307416424473	1.168486997075	86.904872192091
21	(3, 4, 9)	181.055515267153	1.177457009974	88.244262238133
22	(1, 1, 9)	182.052821536235	1.180695443828	88.730337218013
23	(2, 7, 9)	183.427574221835	1.185145008096	89.400375003505
24	(3, 5, 9)	183.522148373539	1.185450495536	89.446469298017
25	(0, 1, 9)	184.455111044520	1.188459884392	89.901183988564

12. Deterministic field spectrum



No direction-specific coefficient is introduced; only a universal field law remains to be declared.

Figure 12.1. Exact path from locked geometry to a deterministic field spectrum.

The shell fixes a finite ordered sequence of reciprocal magnitudes. A universal field equation converts kappa to frequency, energy, decay rate, or mass shift. The geometry does not require a separate coefficient for any address. Examples include a linear wave $\omega_s = c_{\text{field}} \kappa_s$, a quadratic law $E_s = \gamma \kappa_s^2$, and a massive wave $\omega_s^2 = \omega_0^2 + c_{\text{field}}^2 \kappa_s^2$.

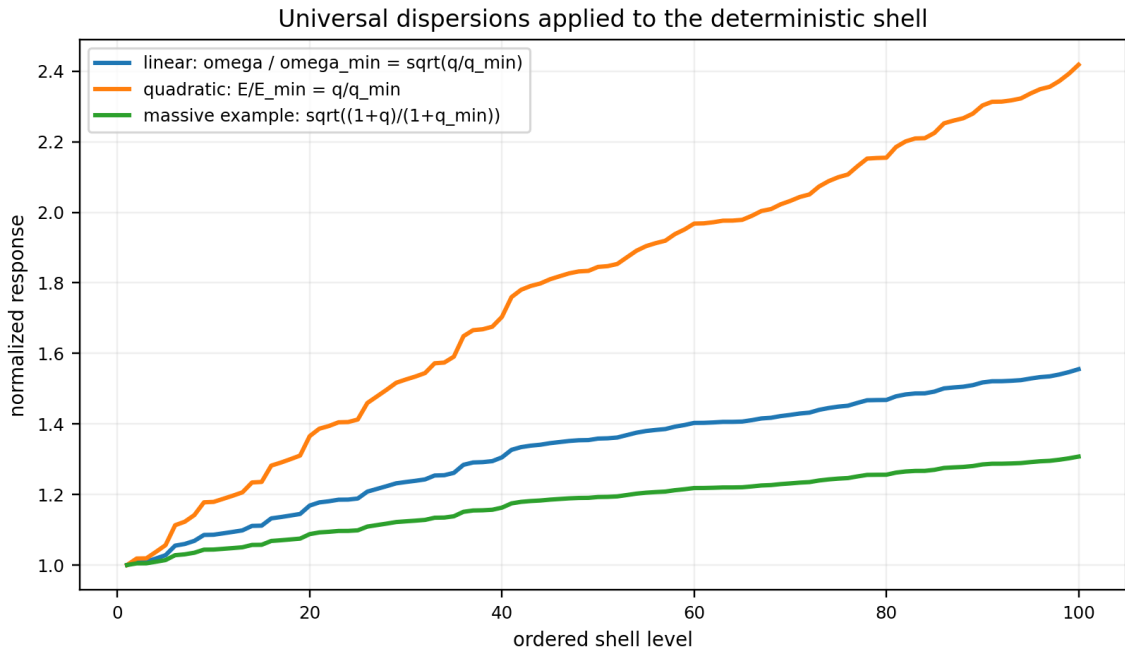


Figure 12.2. Three universal dispersion families applied to the same first 100 deterministic shell levels. Curves are normalized to their lowest level.

linear: $\omega_s / \omega_{\min} = \sqrt{q_s / q_{\min}}$
quadratic: $E_s / E_{\min} = q_s / q_{\min}$
massive: $\omega_s^2 = \omega_0^2 + c_{\text{field}}^2 (2 \pi / 9)^2 q_s$

All relative values are fixed after the universal law is selected. The field spectrum is therefore deterministic. The choice of universal dynamics changes the mapping from q to the measured observable but does not reorder the q levels when the dispersion is monotone.

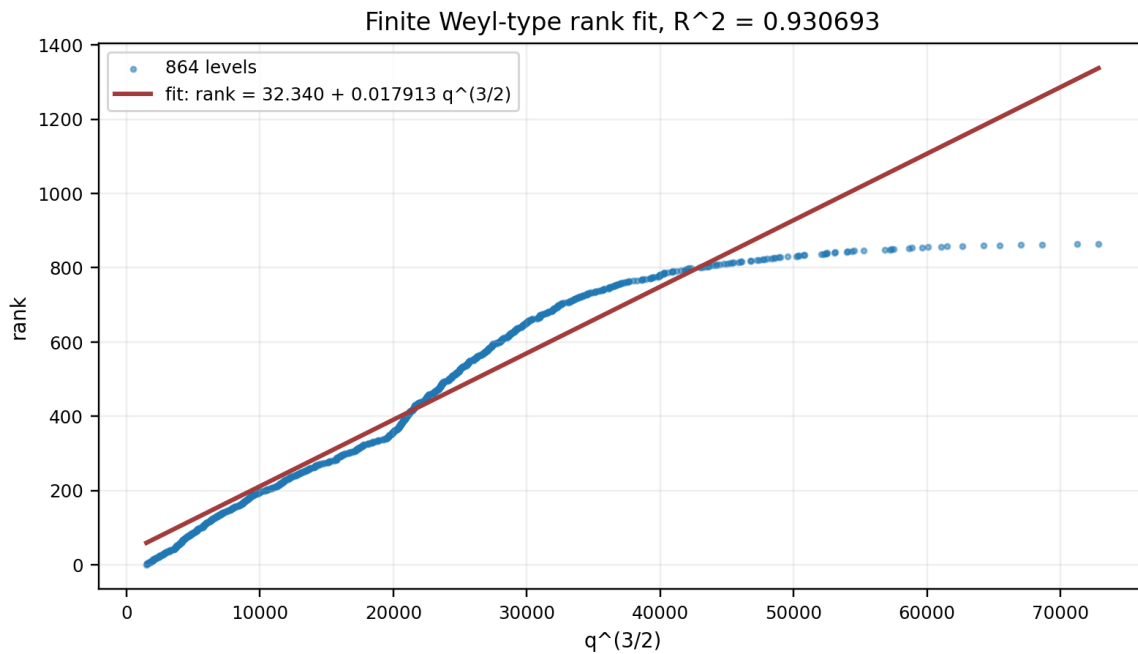


Figure 12.3. Finite Weyl-type rank fit. The R^2 value characterizes broad spectral growth but the residuals retain substantial finite-shell anisotropic structure.

13. Requirements for a particle interpretation

A particle spectrum is more constrained than a field-mode spectrum. Geometry supplies allowed reciprocal directions and form-degree multiplets, but a particle interpretation must specify what physical quantity occupies each mathematical sector. The following choices must be declared before comparison to measured particle data.

Form degree. A scalar particle candidate would occupy $k=0$, a vector candidate $k=1$, a flux or antisymmetric-tensor candidate $k=2$, and a volume pseudoscalar candidate $k=3$. The degree multiplicities are fixed by active directional rank.

Boundary condition and gauge rule. The finite open Hodge problem and the periodic reciprocal shell are distinct spectral problems. A particle grammar must state which one applies and how exact/gauge modes are removed.

Dispersion. A linear, quadratic, or massive mapping must be universal across the entire set. Object-specific scale factors or nearest-level reassignment would destroy determinism.

Address grammar. Charges, spin labels, parity, families, and degeneracies require a predeclared mapping from shell address, form degree, orientation, and symmetry orbit to physical quantum numbers.

The present calculation establishes a finite coefficient-free candidate spectrum. It does not authorize post hoc assignment of known particles to nearby levels.

14. Emergent relations exposed by the Hodge program

14.1 One spectral generator for all form degrees

The same quadratic symbol $\Lambda(j)=\mathbf{x}^T \mathbf{R} \mathbf{x}$ generates all four k -form spectra. Scalar, vector, tensor, and volume behavior differ through exterior multiplicities and exact/coexact placement rather than through separate metric coefficients. This is a unified geometric origin for multiple field sectors.

14.2 Positive circulation without topological circulation

The 1,216 center-graph cycles do not disappear. They move from graph homology into the coexact Hodge sector. This distinguishes dynamical circulation from topological circulation: the former has positive energy and is generated by face curls; the latter would be harmonic and is absent.

14.3 Orientation as a null-mode lifting mechanism

Unsigned laminar adjacency creates 55 zero modes. Cellular orientation couples neighboring dimensions with signs and lifts all 55. Orientation therefore acts as a deterministic mass- or stiffness-generating operation for the unsigned nullspace, leaving only the constant scalar required by connectedness.

14.4 Pascal multiplets and finite supersymmetry

The 1+1, 1+2+1, and 1+3+3+1 modal multiplets are the exterior-algebra dimensions of one-, two-, and three-directional active spaces. Exact and coexact partners share eigenvalues, and the alternating multiplicity vanishes for every positive block. The Supergroup therefore exhibits a finite combinatorial supersymmetry controlled by the cellular differential.

14.5 Open-complex and periodic-shell complementarity

The open cellular Hodge spectra resolve boundary-compatible standing modes on the filled Supergroup. The periodic primitive shell resolves translation-compatible reciprocal directions. They are complementary views of the same locked metric: the first emphasizes exact/coexact topology and form degree, while the second emphasizes anisotropic direction and field dispersion.

15. Worked calculations

15.1 First scalar-vector eigenvalue

For $j=(0,0,1)$, $\mathbf{x}_i=(0,0,2 \sin(\pi/20))$. Because only the c direction is active, $\Lambda = \mathbf{x}_i^T \mathbf{R}_{cc} \mathbf{x}_i$. Numerically $\mathbf{x}_i^T = 0.312868930080$ and $\mathbf{R}_{cc} = 2.731171274712 \text{ nm}^{-2}$, giving $\Lambda = 0.267346073558 \text{ nm}^{-2}$. Active rank $r=1$, so the level occurs once in $k=0$ and once in $k=1$.

$$\Lambda(0,0,1) = (0.312868930080)^2 (2.731171274712) = 0.267346073558 \text{ nm}^{-2}$$

15.2 First bivector level

For $j=(0,1,1)$, two primitive directions are active. The cross term $\mathbf{R}_{bc}$ lowers the level relative to the sum of the separate b and c contributions. Active rank $r=2$ gives multiplicities $(1,2,1,0)$, so this is the first level present in the bivector sector.

$$\Lambda(0,1,1) = \mathbf{x}_i^T (\mathbf{R}_{bb} + 2 \mathbf{R}_{bc} + \mathbf{R}_{cc}) \mathbf{x}_i = 0.294073830399 \text{ nm}^{-2}$$

15.3 First volume-form level

A three-form requires all three primitive directions. The first such mode is $j=(1,1,1)$, with multiplicities $(1,3,3,1)$. Its eigenvalue is $1.069408999171 \text{ nm}^{-2}$.

15.4 A primitive reciprocal shell level

The lowest primitive shell address is $(1, 4, 9)$. Substituting it into $q = \mathbf{s}^T \mathbf{R} \mathbf{s}$ gives $130.593523477839 \text{ nm}^{-2}$. Periodic 9B scaling multiplies by $(2 \pi/9)^2$, giving $\kappa^2 = 63.649699460279 \text{ nm}^{-2}$.

16. Reproducibility and computational implementation

The boundary matrices are stored as sparse NPZ arrays. Their shapes, ranks, nonzero counts, and chain products are recorded in the machine-readable companion. The complete primal and dual modal records and all 864 primitive reciprocal levels are included in that file. Recalculation requires only integer incidence, the locked 3×3 reciprocal metric, and elementary trigonometric symbols.

16.1 Minimal reproduction sequence

First construct the oriented cubical boundary matrices for a $9 \times 9 \times 9$ cell block. Second verify $\mathbf{B}_1 \mathbf{B}_2 = 0$ and $\mathbf{B}_2 \mathbf{B}_3 = 0$. Third compute the locked metric Hodge stars. Fourth enumerate j in $\{0, \dots, 9\}^3$ and evaluate $\Lambda(j) = \mathbf{x}_i(j)^T \mathbf{R} \mathbf{x}_i(j)$. Fifth assign multiplicities $\text{binomial}(r, k)$. Sixth enumerate primitive max-norm-9 integer directions modulo antipodes and evaluate $q(\mathbf{s}) = \mathbf{s}^T \mathbf{R} \mathbf{s}$.

16.2 Operator manifest

File	Bytes	SHA-256
dual8_B1.npz	8,330	f20bfad449164f9adcedfe5e5796022a88f11cebe492b4cb965be94a63954462
dual8_B2.npz	15,358	d5860bc5334047a772ea5d5d247d4352334eff5130875acf779a7f1e474e9abd
dual8_B3.npz	7,038	7866a312d31b71cf36d3802d2efe1ab2e39f2c694a2c2672b6dc4675f2862d06
primal9_B1.npz	11,332	a6ff153c80b06b846912d770dc7ee274a599a4db2660e791162cd912a1c868c5
primal9_B2.npz	21,192	9275e1665aaa1899f1b64f6149b4f3d48703061cb99f9f9e0d04250f21e677ae
primal9_B3.npz	9,779	d356f0b3c36ba1e71dd62766ef886c4e9a8e0c2876293e1c6076173f202673cd

The companion ZIP archives these sparse matrices and the reproducibility code. The machine-readable JSON emitted with this report contains the complete numerical results.

17. Conclusions

The Lucas Supergroup supports a complete metric-weighted cellular Hodge theory with no unresolved topological sector. Its f-vector is (1000,2700,2430,729), its Betti numbers are (1,0,0,0), and its Hodge kernels are one-dimensional only in the scalar sector. Every positive eigenvalue in every form degree descends from the locked reciprocal metric through the same separable quadratic symbol.

The 1,216 center-graph circulation modes are exactly face boundaries in the dual complex. They remain physical finite-energy circulation modes but none is harmonic. The 55 unsigned laminar adjacency modes are distributed across scalar, vector, and bivector cellular sectors; all are lifted by correct oriented incidence, leaving the constant scalar as the unique Hodge harmonic.

The 864 primitive antipodal directions produce 864 distinct reciprocal quadratic values and therefore a finite deterministic field spectrum. No direction-specific or object-specific coefficient is required. The remaining freedom belongs to the universal field equation: the form degree, boundary condition, gauge rule, and dispersion law. A particle interpretation becomes mathematically testable only after those rules are fixed in advance.

Final theorem: $H^0 = R$ and $H^1 = H^2 = H^3 = 0$; the positive spectrum is $\Lambda(j) = x_i(j)^T R x_i(j)$ with multiplicity $\text{binomial}(r(j), k)$.

Appendix A. Complete Cell3 node atlas

Address	lambda	polarity	kind	x (nm)	y (nm)	z (nm)
(-1, -1, -1)	-3	-1	corner	-0.176351350136	0.118682429905	0.637311498325
(-1, -1, 0)	-2	1	edge center	-0.181416008950	-0.076211421278	0.292444294142
(-1, -1, 1)	-1	-1	corner	-0.186480667763	-0.271105272461	-0.052422910042
(-1, 0, -1)	-2	1	edge center	-0.158685962215	0.222342355125	0.363610955631
(-1, 0, 0)	-1	-1	face center	-0.163750621028	0.027448503942	0.018743751447
(-1, 0, 1)	0	1	edge center	-0.168815279841	-0.167445347241	-0.326123452736
(-1, 1, -1)	-1	-1	corner	-0.141020574293	0.326002280345	0.089910412937
(-1, 1, 0)	0	1	edge center	-0.146085233106	0.131108429162	-0.254956791247
(-1, 1, 1)	1	-1	corner	-0.151149891919	-0.063785422021	-0.599823995431
(0, -1, -1)	-2	1	edge center	-0.012600729109	0.091233925963	0.618567746878
(0, -1, 0)	-1	-1	face center	-0.017665387922	-0.103659925220	0.273700542694
(0, -1, 1)	0	1	edge center	-0.022730046735	-0.298553776403	-0.071166661489
(0, 0, -1)	-1	-1	face center	0.005064658813	0.194893851183	0.344867204184
(0, 0, 0)	0	1	body center	0.000000000000	0.000000000000	0.000000000000
(0, 0, 1)	1	-1	face center	-0.005064658813	-0.194893851183	-0.344867204184
(0, 1, -1)	0	1	edge center	0.022730046735	0.298553776403	0.071166661489
(0, 1, 0)	1	-1	face center	0.017665387922	0.103659925220	-0.273700542694
(0, 1, 1)	2	1	edge center	0.012600729109	-0.091233925963	-0.618567746878
(1, -1, -1)	-1	-1	corner	0.151149891919	0.063785422021	0.599823995431
(1, -1, 0)	0	1	edge center	0.146085233106	-0.131108429162	0.254956791247
(1, -1, 1)	1	-1	corner	0.141020574293	-0.326002280345	-0.089910412937
(1, 0, -1)	0	1	edge center	0.168815279841	0.167445347241	0.326123452736
(1, 0, 0)	1	-1	face center	0.163750621028	-0.027448503942	-0.018743751447
(1, 0, 1)	2	1	edge center	0.158685962215	-0.222342355125	-0.363610955631
(1, 1, -1)	1	-1	corner	0.186480667763	0.271105272461	0.052422910042
(1, 1, 0)	2	1	edge center	0.181416008950	0.076211421278	-0.292444294142
(1, 1, 1)	3	-1	corner	0.176351350136	-0.118682429905	-0.637311498325

Appendix B. Ordered primal modal spectrum index

The universal formula and machine-readable companion contain all 1,000 index triples. This printed index contains the lowest 120 and highest 30 modal triples, sufficient to show the complete ordering at both spectral ends.

j	xi_a	xi_b	xi_c	r	Lambda	m0	m1	m2	m3
(0, 0, 0)	0.000000000	0.000000000	0.000000000	0	0.000000000000	1	0	0	0
(0, 0, 1)	0.000000000	0.000000000	0.312868930	1	0.267346073558	1	1	0	0
(0, 1, 1)	0.000000000	0.312868930	0.312868930	2	0.294073830399	1	2	1	0
(0, 1, 0)	0.000000000	0.312868930	0.000000000	1	0.479978790800	1	1	0	0
(0, 1, 2)	0.000000000	0.312868930	0.618033989	2	0.627851865434	1	2	1	0
(1, 0, 0)	0.312868930	0.000000000	0.000000000	1	0.901345694910	1	1	0	0
(1, 0, 1)	0.312868930	0.000000000	0.312868930	2	1.042380768020	1	2	1	0
(0, 0, 2)	0.000000000	0.000000000	0.618033989	1	1.043214597842	1	1	0	0
(1, 1, 1)	0.312868930	0.312868930	0.312868930	3	1.069408999171	1	3	3	1
(0, 2, 2)	0.000000000	0.618033989	0.618033989	2	1.147509326142	1	2	1	0
(0, 2, 1)	0.000000000	0.618033989	0.312868930	2	1.244936045296	1	2	1	0
(1, 1, 2)	0.312868930	0.312868930	0.618033989	3	1.279986229793	1	3	3	1
(1, 1, 0)	0.312868930	0.312868930	0.000000000	2	1.381624960020	1	2	1	0
(0, 1, 3)	0.000000000	0.312868930	0.907980999	2	1.416251214015	1	2	1	0
(0, 2, 3)	0.000000000	0.618033989	0.907980999	2	1.526207347727	1	2	1	0
(1, 0, 2)	0.312868930	0.000000000	0.618033989	2	1.695048487891	1	2	1	0
(1, 2, 2)	0.312868930	0.618033989	0.618033989	3	1.799936766136	1	3	3	1
(0, 2, 0)	0.000000000	0.618033989	0.000000000	1	1.872931494946	1	1	0	0
(1, 1, 3)	0.312868930	0.312868930	0.907980999	3	1.951328582677	1	3	3	1
(1, 2, 1)	0.312868930	0.618033989	0.312868930	3	2.020564289704	1	3	3	1
(1, 2, 3)	0.312868930	0.618033989	0.907980999	3	2.061577792026	1	3	3	1
(0, 0, 3)	0.000000000	0.000000000	0.907980999	1	2.251658155903	1	1	0	0
(0, 2, 4)	0.000000000	0.618033989	1.175570505	2	2.283170323744	1	2	1	0
(0, 3, 3)	0.000000000	0.907980999	0.907980999	2	2.476766274672	1	2	1	0
(0, 3, 2)	0.000000000	0.907980999	0.618033989	2	2.487338491206	1	2	1	0
(0, 1, 4)	0.000000000	0.312868930	1.175570505	2	2.551323883441	1	2	1	0
(1, 2, 4)	0.312868930	0.618033989	1.175570505	3	2.710509912787	1	3	3	1
(1, 2, 0)	0.312868930	0.618033989	0.000000000	2	2.774870739802	1	2	1	0
(1, 0, 3)	0.312868930	0.000000000	0.907980999	2	2.786435050254	1	2	1	0
(0, 3, 4)	0.000000000	0.907980999	1.175570505	2	2.874475226620	1	2	1	0
(1, 1, 4)	0.312868930	0.312868930	1.175570505	3	2.978370396849	1	3	3	1
(0, 3, 1)	0.000000000	0.907980999	0.312868930	2	2.994466537355	1	2	1	0
(1, 3, 3)	0.312868930	0.907980999	0.907980999	3	3.012415179436	1	3	3	1
(1, 3, 2)	0.312868930	0.907980999	0.618033989	3	3.140044391667	1	3	3	1
(0, 2, 5)	0.000000000	0.618033989	1.414213562	2	3.288198725159	1	2	1	0
(1, 3, 4)	0.312868930	0.907980999	1.175570505	3	3.302093276130	1	3	3	1
(2, 0, 0)	0.618033989	0.000000000	0.000000000	1	3.517152782976	1	1	0	0
(2, 0, 1)	0.618033989	0.000000000	0.312868930	2	3.534987051672	1	2	1	0
(0, 3, 5)	0.000000000	0.907980999	1.414213562	2	3.559111844388	1	2	1	0
(2, 1, 1)	0.618033989	0.312868930	0.312868930	3	3.562308358459	1	3	3	1
(1, 2, 5)	0.312868930	0.618033989	1.414213562	3	3.619193677580	1	3	3	1
(2, 1, 2)	0.618033989	0.312868930	0.618033989	3	3.652718397351	1	3	3	1
(1, 3, 1)	0.312868930	0.907980999	0.312868930	3	3.770373242230	1	3	3	1
(0, 0, 4)	0.000000000	0.000000000	1.175570505	1	3.774385872554	1	1	0	0
(1, 3, 5)	0.312868930	0.907980999	1.414213562	3	3.890385257275	1	3	3	1
(0, 1, 5)	0.000000000	0.312868930	1.414213562	2	3.893560029154	1	2	1	0
(2, 1, 0)	0.618033989	0.312868930	0.000000000	2	3.997725123722	1	2	1	0
(0, 3, 0)	0.000000000	0.907980999	0.000000000	1	4.042506196485	1	1	0	0
(2, 0, 2)	0.618033989	0.000000000	0.618033989	2	4.067487579813	1	2	1	0
(0, 4, 3)	0.000000000	1.175570505	0.907980999	2	4.085571120798	1	2	1	0
(0, 4, 4)	0.000000000	1.175570505	1.175570505	2	4.151727744389	1	2	1	0
(2, 2, 2)	0.618033989	0.618033989	0.618033989	3	4.172954792836	1	3	3	1
(1, 0, 4)	0.312868930	0.000000000	1.175570505	2	4.201131911652	1	2	1	0

j	xi_a	xi_b	xi_c	r	Lambda	m0	m1	m2	m3
(2, 1, 3)	0.618033989	0.312868930	0.907980999	3	4.209886086263	1	3	3	1
(1, 1, 5)	0.312868930	0.312868930	1.414213562	3	4.224261905939	1	3	3	1
(2, 2, 3)	0.618033989	0.618033989	0.907980999	3	4.320421154753	1	3	3	1
(0, 3, 6)	0.000000000	0.907980999	1.618033989	2	4.390152237440	1	2	1	0
(0, 2, 6)	0.000000000	0.618033989	1.618033989	2	4.392879552406	1	2	1	0
(0, 4, 2)	0.000000000	1.175570505	0.618033989	2	4.455397361400	1	2	1	0
(2, 2, 1)	0.618033989	0.618033989	0.312868930	3	4.513749508133	1	3	3	1
(0, 4, 5)	0.000000000	1.175570505	1.414213562	2	4.540677648192	1	2	1	0
(1, 4, 4)	0.312868930	1.175570505	1.175570505	3	4.579602782575	1	3	3	1
(1, 4, 3)	0.312868930	1.175570505	0.907980999	3	4.621477014238	1	3	3	1
(1, 3, 6)	0.312868930	0.907980999	1.618033989	3	4.639139557041	1	3	3	1
(1, 2, 6)	0.312868930	0.618033989	1.618033989	3	4.641588411540	1	3	3	1
(2, 2, 4)	0.618033989	0.618033989	1.175570505	3	4.863982498450	1	3	3	1
(1, 4, 5)	0.312868930	1.175570505	1.414213562	3	4.872208049756	1	3	3	1
(1, 3, 0)	0.312868930	0.907980999	0.000000000	2	4.944723901807	1	2	1	0
(2, 0, 3)	0.618033989	0.000000000	0.907980999	2	5.044699478205	1	2	1	0
(1, 4, 2)	0.312868930	1.175570505	0.618033989	3	5.108360250536	1	3	3	1
(0, 4, 6)	0.000000000	1.175570505	1.618033989	2	5.119177729900	1	2	1	0
(2, 1, 4)	0.618033989	0.312868930	1.175570505	3	5.131557123371	1	3	3	1
(0, 3, 7)	0.000000000	0.907980999	1.782013048	2	5.223467568409	1	2	1	0
(2, 3, 3)	0.618033989	0.907980999	0.907980999	3	5.271530146009	1	3	3	1
(0, 1, 6)	0.000000000	0.312868930	1.618033989	2	5.286243477795	1	2	1	0
(0, 4, 1)	0.000000000	1.175570505	0.312868930	2	5.340635100959	1	2	1	0
(1, 4, 6)	0.312868930	1.175570505	1.618033989	3	5.368422038176	1	3	3	1
(2, 2, 0)	0.618033989	0.618033989	0.000000000	2	5.391256762645	1	2	1	0
(1, 3, 7)	0.312868930	0.907980999	1.782013048	3	5.406253494766	1	3	3	1
(0, 2, 7)	0.000000000	0.618033989	1.782013048	2	5.446346032466	1	2	1	0
(2, 3, 4)	0.618033989	0.907980999	1.175570505	3	5.455837465639	1	3	3	1
(0, 0, 5)	0.000000000	0.000000000	1.414213562	1	5.462342549423	1	1	0	0
(2, 3, 2)	0.618033989	0.907980999	0.618033989	3	5.513334022212	1	3	3	1
(1, 1, 6)	0.312868930	0.312868930	1.618033989	3	5.534659261293	1	3	3	1
(1, 2, 7)	0.312868930	0.618033989	1.782013048	3	5.628853498357	1	3	3	1
(2, 2, 5)	0.618033989	0.618033989	1.414213562	3	5.678693951324	1	3	3	1
(0, 4, 7)	0.000000000	1.175570505	1.782013048	2	5.749317532742	1	2	1	0
(1, 0, 5)	0.312868930	0.000000000	1.414213562	2	5.792743951898	1	2	1	0
(0, 5, 4)	0.000000000	1.414213562	1.175570505	2	5.883187788745	1	2	1	0
(0, 3, 8)	0.000000000	0.907980999	1.902113033	2	5.926978259886	1	2	1	0
(1, 4, 7)	0.312868930	1.175570505	1.782013048	3	5.932360447775	1	3	3	1
(2, 3, 5)	0.618033989	0.907980999	1.414213562	3	5.950157134864	1	3	3	1
(0, 5, 5)	0.000000000	1.414213562	1.414213562	2	6.008436836494	1	2	1	0
(1, 3, 8)	0.312868930	0.907980999	1.902113033	3	6.061277591055	1	3	3	1
(0, 5, 3)	0.000000000	1.414213562	0.907980999	2	6.112717879118	1	2	1	0
(1, 4, 1)	0.312868930	1.175570505	0.312868930	3	6.116798794509	1	3	3	1
(2, 3, 1)	0.618033989	0.907980999	0.312868930	3	6.263830064504	1	3	3	1
(2, 1, 5)	0.618033989	0.312868930	1.414213562	3	6.283476320542	1	3	3	1
(0, 4, 8)	0.000000000	1.175570505	1.902113033	2	6.304020335112	1	2	1	0
(0, 2, 8)	0.000000000	0.618033989	1.902113033	2	6.311097736000	1	2	1	0
(1, 5, 4)	0.312868930	1.414213562	1.175570505	3	6.311292015902	1	3	3	1
(1, 5, 5)	0.312868930	1.414213562	1.414213562	3	6.340196427029	1	3	3	1
(2, 0, 4)	0.618033989	0.000000000	1.175570505	2	6.354025562538	1	2	1	0
(0, 5, 6)	0.000000000	1.414213562	1.618033989	2	6.361715110305	1	2	1	0
(0, 3, 9)	0.000000000	0.907980999	1.975376681	2	6.394826559637	1	2	1	0
(1, 4, 8)	0.312868930	1.175570505	1.902113033	3	6.438576654957	1	3	3	1
(1, 2, 8)	0.312868930	0.618033989	1.902113033	3	6.445118606702	1	3	3	1
(1, 3, 9)	0.312868930	0.907980999	1.975376681	3	6.499547994564	1	3	3	1
(0, 1, 7)	0.000000000	0.312868930	1.782013048	2	6.571415875980	1	2	1	0
(1, 5, 6)	0.312868930	1.414213562	1.618033989	3	6.611188607552	1	3	3	1
(2, 3, 6)	0.618033989	0.907980999	1.618033989	3	6.618651498051	1	3	3	1

j	xi_a	xi_b	xi_c	r	Lambda	m0	m1	m2	m3
(2, 2, 6)	0.618033989	0.618033989	1.618033989	3	6.620828748705	1	3	3	1
(1, 5, 3)	0.312868930	1.414213562	0.907980999	3	6.648852961530	1	3	3	1
(0, 4, 9)	0.000000000	1.175570505	1.975376681	2	6.681092528873	1	2	1	0
(2, 4, 4)	0.618033989	1.175570505	1.175570505	3	6.733597632845	1	3	3	1
(1, 1, 7)	0.312868930	0.312868930	1.782013048	3	6.753630266235	1	3	3	1
(0, 4, 0)	0.000000000	1.175570505	0.000000000	1	6.776329807314	1	1	0	0
(1, 4, 9)	0.312868930	1.175570505	1.975376681	3	6.786070952476	1	3	3	1
(0, 5, 2)	0.000000000	1.414213562	0.618033989	2	6.802935903368	1	2	1	0
(0, 5, 7)	0.000000000	1.414213562	1.782013048	2	6.810657861787	1	2	1	0
(9, 8, 4)	1.975376681	1.902113033	1.175570505	3	44.107053482808	1	3	3	1
(8, 9, 3)	1.902113033	1.975376681	0.907980999	3	44.178151067458	1	3	3	1
(7, 7, 0)	1.782013048	1.782013048	0.000000000	2	44.821569076441	1	2	1	0
(7, 9, 1)	1.782013048	1.975376681	0.312868930	3	45.071354088281	1	3	3	1
(9, 9, 4)	1.975376681	1.975376681	1.175570505	3	45.101654099784	1	3	3	1
(8, 8, 2)	1.902113033	1.902113033	0.618033989	3	45.149699312615	1	3	3	1
(9, 8, 3)	1.975376681	1.902113033	0.907980999	3	45.623187277753	1	3	3	1
(9, 5, 0)	1.975376681	1.414213562	0.000000000	2	45.746128309576	1	2	1	0
(8, 7, 1)	1.902113033	1.782013048	0.312868930	3	45.814278932155	1	3	3	1
(9, 7, 2)	1.975376681	1.782013048	0.618033989	3	45.880895592845	1	3	3	1
(9, 6, 1)	1.975376681	1.618033989	0.312868930	3	45.903646281671	1	3	3	1
(8, 6, 0)	1.902113033	1.618033989	0.000000000	2	46.161660049345	1	2	1	0
(8, 9, 2)	1.902113033	1.975376681	0.618033989	3	46.333420145465	1	3	3	1
(9, 9, 3)	1.975376681	1.975376681	0.907980999	3	46.708564000719	1	3	3	1
(7, 8, 0)	1.782013048	1.902113033	0.000000000	2	46.991800734010	1	2	1	0
(9, 8, 2)	1.975376681	1.902113033	0.618033989	3	47.707506681226	1	3	3	1
(8, 8, 1)	1.902113033	1.902113033	0.312868930	3	47.810566857361	1	3	3	1
(7, 9, 0)	1.782013048	1.975376681	0.000000000	2	48.385154195875	1	2	1	0
(9, 7, 1)	1.975376681	1.782013048	0.312868930	3	48.400908881699	1	3	3	1
(9, 6, 0)	1.975376681	1.618033989	0.000000000	2	48.777831017805	1	2	1	0
(9, 9, 2)	1.975376681	1.975376681	0.618033989	3	48.891243990371	1	3	3	1
(8, 7, 0)	1.902113033	1.782013048	0.000000000	2	48.896441090514	1	2	1	0
(8, 9, 1)	1.902113033	1.975376681	0.312868930	3	49.097810792506	1	3	3	1
(9, 8, 1)	1.975376681	1.902113033	0.312868930	3	50.397223816245	1	3	3	1
(8, 8, 0)	1.902113033	1.902113033	0.000000000	2	51.066717024083	1	2	1	0
(9, 7, 0)	1.975376681	1.782013048	0.000000000	2	51.512648936300	1	2	1	0
(9, 9, 1)	1.975376681	1.975376681	0.312868930	3	51.684484227685	1	3	3	1
(8, 9, 0)	1.902113033	1.975376681	0.000000000	2	52.460097495288	1	2	1	0
(9, 8, 0)	1.975376681	1.902113033	0.000000000	2	53.682951879208	1	2	1	0
(9, 9, 0)	1.975376681	1.975376681	0.000000000	2	55.076348826708	1	2	1	0

Appendix C. Ordered dual center-complex modal spectrum index

j	xi_a	xi_b	xi_c	r	Lambda	m0	m1	m2	m3
(0, 0, 0)	0.000000000	0.000000000	0.000000000	0	0.000000000000	1	0	0	0
(0, 0, 1)	0.000000000	0.000000000	0.347296355	1	0.329419563525	1	1	0	0
(0, 1, 1)	0.000000000	0.347296355	0.347296355	2	0.362353078782	1	2	1	0
(0, 1, 0)	0.000000000	0.347296355	0.000000000	1	0.591422202923	1	1	0	0
(0, 1, 2)	0.000000000	0.347296355	0.684040287	2	0.769359616782	1	2	1	0
(1, 0, 0)	0.347296355	0.000000000	0.000000000	1	1.110623774836	1	1	0	0
(0, 0, 2)	0.000000000	0.000000000	0.684040287	1	1.277945393025	1	1	0	0
(1, 0, 1)	0.347296355	0.000000000	0.347296355	2	1.284404940228	1	2	1	0
(1, 1, 1)	0.347296355	0.347296355	0.347296355	3	1.317708695133	1	3	3	1
(0, 2, 2)	0.000000000	0.684040287	0.684040287	2	1.405707186066	1	2	1	0
(0, 2, 1)	0.000000000	0.684040287	0.347296355	2	1.523766149917	1	2	1	0
(1, 1, 2)	0.347296355	0.347296355	0.684040287	3	1.573805828970	1	3	3	1
(1, 1, 0)	0.347296355	0.347296355	0.000000000	2	1.702416217408	1	2	1	0
(0, 1, 3)	0.000000000	0.347296355	1.000000000	2	1.714489392784	1	2	1	0
(0, 2, 3)	0.000000000	0.684040287	1.000000000	2	1.858179099446	1	2	1	0
(1, 0, 2)	0.347296355	0.000000000	0.684040287	2	2.082021365565	1	2	1	0
(1, 2, 2)	0.347296355	0.684040287	0.684040287	3	2.210512388358	1	3	3	1
(0, 2, 0)	0.000000000	0.684040287	0.000000000	1	2.294354565559	1	1	0	0
(1, 1, 3)	0.347296355	0.347296355	1.000000000	3	2.377340500661	1	3	3	1
(1, 2, 1)	0.347296355	0.684040287	0.347296355	3	2.479480756372	1	3	3	1
(1, 2, 3)	0.347296355	0.684040287	1.000000000	3	2.521389197427	1	3	3	1
(0, 0, 3)	0.000000000	0.000000000	1.000000000	1	2.731171274712	1	1	0	0
(0, 2, 4)	0.000000000	0.684040287	1.285575219	2	2.736308804316	1	2	1	0
(0, 3, 3)	0.000000000	1.000000000	1.000000000	2	3.004218418247	1	2	1	0
(0, 3, 2)	0.000000000	1.000000000	0.684040287	2	3.013996964569	1	2	1	0
(0, 1, 4)	0.000000000	0.347296355	1.285575219	2	3.037900161189	1	2	1	0
(1, 2, 4)	0.347296355	0.684040287	1.285575219	3	3.271540393432	1	3	3	1
(1, 0, 3)	0.347296355	0.000000000	1.000000000	2	3.393652142941	1	2	1	0
(1, 2, 0)	0.347296355	0.684040287	0.000000000	2	3.405707570147	1	2	1	0
(0, 3, 4)	0.000000000	1.000000000	1.285575219	2	3.464550312645	1	2	1	0
(1, 1, 4)	0.347296355	0.347296355	1.285575219	3	3.572772760201	1	3	3	1
(0, 3, 1)	0.000000000	1.000000000	0.347296355	2	3.624713791042	1	2	1	0
(1, 3, 3)	0.347296355	1.000000000	1.000000000	3	3.667765349055	1	3	3	1
(1, 3, 2)	0.347296355	1.000000000	0.684040287	3	3.819138999687	1	3	3	1
(0, 2, 5)	0.000000000	0.684040287	1.532088886	2	3.852566204003	1	2	1	0
(1, 3, 4)	0.347296355	1.000000000	1.285575219	3	4.000118734588	1	3	3	1
(0, 3, 5)	0.000000000	1.000000000	1.532088886	2	4.220157126144	1	2	1	0
(1, 2, 5)	0.347296355	0.684040287	1.532088886	3	4.277324441932	1	3	3	1
(2, 0, 0)	0.684040287	0.000000000	0.000000000	1	4.308537481038	1	1	0	0
(2, 0, 1)	0.684040287	0.000000000	0.347296355	2	4.331409242267	1	2	1	0
(2, 1, 1)	0.684040287	0.347296355	0.347296355	3	4.365071987276	1	3	3	1
(2, 1, 2)	0.684040287	0.347296355	0.684040287	3	4.474845022831	1	3	3	1
(0, 0, 4)	0.000000000	0.000000000	1.285575219	1	4.513816719923	1	1	0	0
(0, 1, 5)	0.000000000	0.347296355	1.532088886	2	4.538532185532	1	2	1	0
(1, 3, 1)	0.347296355	1.000000000	0.347296355	3	4.580765230323	1	3	3	1
(1, 3, 5)	0.347296355	1.000000000	1.532088886	3	4.645252196899	1	3	3	1
(0, 4, 3)	0.000000000	1.285575219	1.000000000	2	4.882370817165	1	2	1	0
(2, 1, 0)	0.684040287	0.347296355	0.000000000	2	4.900688913713	1	2	1	0
(0, 3, 0)	0.000000000	1.000000000	0.000000000	1	4.903398312368	1	1	0	0
(1, 1, 5)	0.347296355	0.347296355	1.532088886	3	4.962931433357	1	3	3	1
(0, 4, 4)	0.000000000	1.285575219	1.285575219	2	4.965082729210	1	2	1	0
(2, 0, 2)	0.684040287	0.000000000	0.684040287	2	4.982701569322	1	2	1	0
(0, 2, 6)	0.000000000	0.684040287	1.732050808	2	5.001862909398	1	2	1	0
(1, 0, 4)	0.347296355	0.000000000	1.285575219	2	5.048319079288	1	2	1	0
(0, 3, 6)	0.000000000	1.000000000	1.732050808	2	5.076908655199	1	2	1	0

j	xi_a	xi_b	xi_c	r	Lambda	m0	m1	m2	m3
(2, 2, 2)	0.684040287	0.684040287	0.684040287	3	5.111899664591	1	3	3	1
(2, 1, 3)	0.684040287	0.347296355	1.000000000	3	5.141086885805	1	3	3	1
(2, 2, 3)	0.684040287	0.684040287	1.000000000	3	5.285483664943	1	3	3	1
(0, 4, 2)	0.000000000	1.285575219	0.684040287	2	5.309947173959	1	2	1	0
(1, 2, 6)	0.347296355	0.684040287	1.732050808	3	5.337009630691	1	3	3	1
(0, 4, 5)	0.000000000	1.285575219	1.532088886	2	5.394721180552	1	2	1	0
(1, 3, 6)	0.347296355	1.000000000	1.732050808	3	5.412392209318	1	3	3	1
(1, 4, 4)	0.347296355	1.285575219	1.285575219	3	5.500955592208	1	3	3	1
(2, 2, 1)	0.684040287	0.684040287	0.347296355	3	5.527192130886	1	3	3	1
(1, 4, 3)	0.347296355	1.285575219	1.000000000	3	5.546222189028	1	3	3	1
(1, 4, 5)	0.347296355	1.285575219	1.532088886	3	5.820120692363	1	3	3	1
(0, 3, 7)	0.000000000	1.000000000	1.879385242	2	5.847924368190	1	2	1	0
(2, 2, 4)	0.684040287	0.684040287	1.285575219	3	5.911544914315	1	3	3	1
(0, 4, 6)	0.000000000	1.285575219	1.732050808	2	5.987060351385	1	2	1	0
(0, 2, 7)	0.000000000	0.684040287	1.879385242	2	5.988429551734	1	2	1	0
(0, 1, 6)	0.000000000	0.347296355	1.732050808	2	5.999618048237	1	2	1	0
(1, 3, 0)	0.347296355	1.000000000	0.000000000	2	6.015088149783	1	2	1	0
(1, 4, 2)	0.347296355	1.285575219	0.684040287	3	6.115393650133	1	3	3	1
(1, 3, 7)	0.347296355	1.000000000	1.879385242	3	6.117381040813	1	3	3	1
(2, 0, 3)	0.684040287	0.000000000	1.000000000	2	6.157039537981	1	2	1	0
(2, 1, 4)	0.684040287	0.347296355	1.285575219	3	6.212429198713	1	3	3	1
(1, 2, 7)	0.347296355	0.684040287	1.879385242	3	6.257549391529	1	3	3	1
(1, 4, 6)	0.347296355	1.285575219	1.732050808	3	6.322848346559	1	3	3	1
(1, 1, 6)	0.347296355	0.347296355	1.732050808	3	6.334405779426	1	3	3	1
(0, 4, 1)	0.000000000	1.285575219	0.347296355	2	6.365945063968	1	2	1	0
(0, 3, 8)	0.000000000	1.000000000	1.969615506	2	6.378652387218	1	2	1	0
(0, 0, 5)	0.000000000	0.000000000	1.532088886	1	6.410868378923	1	1	0	0
(2, 3, 3)	0.684040287	1.000000000	1.000000000	3	6.432186414902	1	3	3	1
(0, 4, 7)	0.000000000	1.285575219	1.879385242	2	6.563253745793	1	2	1	0
(2, 2, 0)	0.684040287	0.684040287	0.000000000	2	6.604328348824	1	2	1	0
(1, 3, 8)	0.347296355	1.000000000	1.969615506	3	6.607673006865	1	3	3	1
(2, 3, 4)	0.684040287	1.000000000	1.285575219	3	6.640449853802	1	3	3	1
(0, 2, 8)	0.000000000	0.684040287	1.969615506	2	6.651164847196	1	2	1	0
(2, 3, 2)	0.684040287	1.000000000	0.684040287	3	6.720852874252	1	3	3	1
(2, 2, 5)	0.684040287	0.684040287	1.532088886	3	6.810212288500	1	3	3	1
(1, 4, 7)	0.347296355	1.285575219	1.879385242	3	6.833014859470	1	3	3	1
(1, 0, 5)	0.347296355	0.000000000	1.532088886	2	6.834897387100	1	2	1	0
(1, 2, 8)	0.347296355	0.684040287	1.969615506	3	6.879848634016	1	3	3	1
(0, 5, 4)	0.000000000	1.532088886	1.285575219	2	6.903534246824	1	2	1	0
(0, 4, 8)	0.000000000	1.285575219	1.969615506	2	6.974669063441	1	2	1	0
(0, 5, 5)	0.000000000	1.532088886	1.532088886	2	7.051790943778	1	2	1	0
(0, 5, 3)	0.000000000	1.532088886	1.000000000	2	7.146790696936	1	2	1	0
(2, 3, 5)	0.684040287	1.000000000	1.532088886	3	7.178466641800	1	3	3	1
(1, 4, 8)	0.347296355	1.285575219	1.969615506	3	7.203994124143	1	3	3	1
(0, 1, 7)	0.000000000	0.347296355	1.879385242	2	7.215914824764	1	2	1	0
(8, 6, 2)	1.969615506	1.732050808	0.684040287	3	44.495566092370	1	3	3	1
(8, 7, 3)	1.969615506	1.879385242	1.000000000	3	44.539492585680	1	3	3	1
(7, 8, 2)	1.879385242	1.969615506	0.684040287	3	44.937712974511	1	3	3	1
(6, 7, 0)	1.732050808	1.879385242	0.000000000	2	44.953307421545	1	2	1	0
(8, 8, 3)	1.969615506	1.969615506	1.000000000	3	45.825172664728	1	3	3	1
(7, 7, 1)	1.879385242	1.879385242	0.347296355	3	46.318584752655	1	3	3	1
(8, 7, 2)	1.969615506	1.879385242	0.684040287	3	46.638841375172	1	3	3	1
(6, 8, 0)	1.732050808	1.969615506	0.000000000	2	46.656719512471	1	2	1	0
(8, 6, 1)	1.969615506	1.732050808	0.347296355	3	47.103577524583	1	3	3	1
(8, 5, 0)	1.969615506	1.532088886	0.000000000	2	47.240470249873	1	2	1	0
(7, 6, 0)	1.879385242	1.732050808	0.000000000	2	47.243751723155	1	2	1	0
(7, 8, 1)	1.879385242	1.969615506	0.347296355	3	47.876937994142	1	3	3	1
(8, 8, 2)	1.969615506	1.969615506	0.684040287	3	48.056528730654	1	3	3	1

j	xi_a	xi_b	xi_c	r	Lambda	m0	m1	m2	m3
(8, 7, 1)	1.969615506	1.879385242	0.347296355	3	49.476582941576	1	3	3	1
(7, 7, 0)	1.879385242	1.879385242	0.000000000	2	49.853645437904	1	2	1	0
(8, 6, 0)	1.969615506	1.732050808	0.000000000	2	50.442145157648	1	2	1	0
(8, 8, 1)	1.969615506	1.969615506	0.347296355	3	51.034961174252	1	3	3	1
(7, 8, 0)	1.879385242	1.969615506	0.000000000	2	51.557098336234	1	2	1	0
(8, 7, 0)	1.969615506	1.879385242	0.000000000	2	53.052079679800	1	2	1	0
(8, 8, 0)	1.969615506	1.969615506	0.000000000	2	54.755557569320	1	2	1	0

Appendix D. Ordered primitive reciprocal spectrum index

Rank	Address	q (nm ⁻²)	sqrt(q)	q/qmin	sqrt ratio	kappa ²
1	(1, 4, 9)	130.593523477839	11.427752337089	1.000000000000	1.000000000000	63.649699460279
2	(0, 4, 9)	132.986604171555	11.531981797226	1.018324650641	1.009120731449	64.816057966297
3	(1, 5, 9)	133.054017374512	11.534904307124	1.018840856967	1.009376469394	64.848914343808
4	(0, 5, 9)	135.444028463372	11.638042295136	1.037142002577	1.018401690188	66.013776761648
5	(1, 3, 9)	137.939826205902	11.744778678455	1.056253193362	1.027741793138	67.230198311470
6	(1, 6, 9)	145.321307895922	12.054928780209	1.112775764264	1.054881872185	70.827842962057
7	(2, 4, 9)	146.616493843039	12.108529796926	1.122693453232	1.059572297312	71.459100884224
8	(2, 5, 9)	149.080057344568	12.209834451972	1.141557815230	1.068437089973	72.659811855931
9	(1, -4, -9)	153.795735924186	12.401440880970	1.177667405155	1.085203854193	74.958176402278
10	(2, 3, 9)	153.959726966245	12.408050893120	1.178923141563	1.085782271711	75.038103647238
11	(1, 2, 9)	155.092925558701	12.453631019052	1.187600437054	1.089770818592	75.590410897382
12	(1, -5, -9)	156.250090611146	12.500003624445	1.196461252059	1.093828712395	76.154399109451
13	(0, 2, 9)	157.492145462129	12.549587461830	1.205972097758	1.098167609137	76.759761579755
14	(1, -3, -9)	161.148177861961	12.694415223316	1.233967608580	1.110840946571	78.541667429824
15	(2, 6, 9)	161.350417470834	12.702378417873	1.235516227558	1.111537776037	78.640236562358
16	(1, 7, 9)	167.395395042067	12.938137232309	1.281804721889	1.132168150890	81.586485315026
17	(1, -6, -9)	168.511241922843	12.981188001213	1.290349149293	1.135935363167	82.130335551344
18	(0, 7, 9)	169.779266921214	13.029937333741	1.300058857437	1.140201235500	82.748355556510
19	(2, 2, 9)	171.109756714187	13.080892810286	1.310246880223	1.144660159271	83.396820144971
20	(1, -2, -9)	178.307416424473	13.353180011685	1.365361862334	1.168486997075	86.904872192091
21	(3, 4, 9)	181.055515267153	13.455687097549	1.386405010336	1.177457009974	88.244262238133
22	(1, 1, 9)	182.052821536235	13.492695117590	1.394041731075	1.180695443828	88.730337218013
23	(2, 7, 9)	183.427574221835	13.543543636059	1.404568690215	1.185145008096	89.400375003505
24	(3, 5, 9)	183.522148373539	13.547034670862	1.405292877366	1.185450495536	89.446469298017
25	(0, 1, 9)	184.455111044520	13.581425221401	1.412436896810	1.188459884392	89.901183988564
26	(1, -7, -9)	190.579189859275	13.805042189696	1.459331096857	1.208027771558	92.885985727958
27	(2, -4, -9)	193.020918735732	13.893196850823	1.478028263542	1.215741857280	94.076054768222
28	(2, -5, -9)	195.472203817836	13.981137429331	1.496798605415	1.223437209429	95.270781387217
29	(2, 1, 9)	198.066583086865	14.073613007571	1.516664669213	1.231529402496	96.535250377425
30	(1, 8, 9)	199.276278812948	14.116525026116	1.525927729845	1.235284473247	97.124841402715
31	(2, -3, -9)	200.376430278364	14.155438187437	1.534351971998	1.238689618911	97.661041883946
32	(0, 8, 9)	201.657081087238	14.200601433997	1.544158360360	1.242641686232	98.285215556022
33	(1, -1, -9)	205.273451611720	14.327367225409	1.571850166418	1.253734488007	100.047790689078
34	(3, 2, 9)	205.542638928589	14.336758313112	1.573911427265	1.254556267078	100.178989322524
35	(2, -6, -9)	207.730285524676	14.412851401603	1.590662997617	1.261214889548	101.245221740933
36	(2, 8, 9)	215.311527597573	14.673497456216	1.648715203202	1.284023053999	104.940227179372
37	(2, -2, -9)	217.538738445732	14.749194501590	1.665769730783	1.290647020212	106.025742734391
38	(3, 7, 9)	217.875804460519	14.760616669385	1.668350762414	1.291646531530	106.190024621947
39	(1, 0, 9)	218.819514138506	14.792549277880	1.675577075425	1.294440835042	106.649977273365
40	(1, -8, -9)	222.453934420444	14.914889688511	1.703407094749	1.305146388245	108.421349639292
41	(2, -7, -9)	229.795163856252	15.158996136165	1.759621440149	1.326507233357	111.999375829369
42	(3, 1, 9)	232.496395696411	15.247832491748	1.780305711224	1.334280971619	113.315923466800
43	(4, 4, 9)	233.910587750182	15.294135730736	1.791134671314	1.338332795426	114.005183522005
44	(2, 0, 9)	234.830206084280	15.324170649150	1.798176508532	1.340961039155	114.453394344599
45	(4, 5, 9)	236.380290461425	15.374663913771	1.810046043375	1.345379516484	115.208886670067
46	(1, 9, 8)	237.497584129361	15.410956626029	1.818601549331	1.348555356421	115.753442052900
47	(0, 9, 8)	238.584940727395	15.446195024257	1.826927816737	1.351638937267	116.283406470950
48	(0, 9, 7)	239.290532126215	15.469018460336	1.832330775322	1.353636131064	116.627303161082
49	(1, 9, 7)	239.493551599581	15.475579200779	1.833885365994	1.354210236999	116.726252390111
50	(1, 9, 9)	240.963959208565	15.523013857127	1.845144788129	1.358361066922	117.442911225125
51	(4, 3, 9)	241.247681663676	15.532149936943	1.847317349582	1.359160531204	117.581194108663
52	(1, 0, -9)	242.046283423704	15.557836720563	1.853432520831	1.361408285868	117.970422920786
53	(2, -1, -9)	244.507843237835	15.636746568191	1.872281539898	1.368313392428	119.170157319556
54	(1, 9, 6)	246.951861619224	15.714702084966	1.890996238118	1.375134989953	120.361342236756
55	(4, 6, 9)	248.656789797403	15.768855056643	1.904051465765	1.379873713702	121.192303552849

Rank	Address	q (nm^-2)	sqrt(q)	q/qmin	sqrt ratio	kappa^2
56	(3, 8, 9)	249.762827441113	15.803886466345	1.912520780431	1.382939181754	121.731372885992
57	(3, -4, -9)	250.662152606193	15.832313558233	1.919407225801	1.385426730578	122.169693064129
58	(3, -5, -9)	253.110368083441	15.909442733278	1.938154062643	1.392176017120	123.362923594947
59	(2, 9, 8)	254.826278590242	15.963279067605	1.951293385797	1.396887034014	124.199237564814
60	(2, 9, 9)	257.002277598046	16.031290578055	1.967955766517	1.402838467721	125.259793089959
61	(0, 9, 5)	257.088742572124	16.033987107770	1.968617858877	1.403074430982	125.301935069648
62	(1, -9, -7)	257.503563711764	16.046917576649	1.971794288524	1.404205928104	125.504113862016
63	(1, -9, -8)	258.088348384344	16.065128333890	1.976272188017	1.405799483574	125.789130818963
64	(2, 9, 7)	258.112622131862	16.065883795542	1.976458060538	1.405865591206	125.800961549103
65	(4, 2, 9)	258.391572201905	16.074562893028	1.978594078180	1.406625066668	125.936918430041
66	(1, 9, 5)	259.872514188290	16.120561844684	1.989934165705	1.410650263426	126.658711592835
67	(2, -8, -9)	261.666838812564	16.176119399057	2.003673933011	1.415511897870	127.533243652525
68	(1, -9, -6)	262.381121588607	16.198182663145	2.009143444492	1.417442571850	127.881376414503
69	(1, -9, -9)	264.135475606348	16.252245248160	2.022577142971	1.422173387098	128.736427285345
70	(1, -1, 9)	265.393003365513	16.290887126412	2.032206470105	1.425554793793	129.349331063436
71	(2, 9, 6)	266.861308222904	16.335890187648	2.043449790741	1.429492843893	130.064965042827
72	(0, 1, -9)	267.801432083510	16.364639686944	2.050648645903	1.432008605387	130.523170010343
73	(4, 7, 9)	270.740085758118	16.454181406503	2.073150938485	1.439844067420	131.955434170352
74	(1, -9, -5)	272.721022014874	16.514267226095	2.088319655922	1.445101953470	132.920918476424
75	(0, 9, 4)	274.181361619214	16.558422679084	2.099501983846	1.448965832532	133.632670288082
76	(3, -2, -9)	275.186111525906	16.588734476322	2.107195703105	1.451618304895	134.122373206654
77	(1, 9, 4)	278.255509306780	16.680992455690	2.130699148752	1.459691456696	135.618360458348
78	(2, 9, 5)	281.072336863371	16.765212103143	2.152268576405	1.467061204042	136.991248045985
79	(2, 0, -9)	281.283744654675	16.771515872296	2.153887399343	1.467612823378	137.094285639441
80	(2, -1, 9)	281.400625706430	16.775000021056	2.154782398181	1.467917708245	137.151252046493
81	(4, 1, 9)	285.342259364871	16.892076822134	2.184964857107	1.478162662601	139.072356486139
82	(3, -7, -9)	287.427188912144	16.953677740011	2.200929887315	1.483553129253	140.088525860743
83	(1, -9, -4)	288.523264990564	16.985972594778	2.209322922813	1.486379131586	140.622740047780
84	(1, 1, -9)	288.625911860423	16.988993844852	2.210108925573	1.486643509915	140.672768872123
85	(3, 9, 8)	290.571024110039	17.046143966013	2.225003326136	1.491644503941	141.620793006690
86	(2, -9, -7)	294.132646356228	17.150295809584	2.252275905598	1.500758443454	143.356684492913
87	(3, 9, 7)	295.147743723058	17.179864485003	2.260048858955	1.503345887996	143.851430638059
88	(2, -9, -8)	296.007807100209	17.204877421830	2.266634663169	1.505534676840	144.270615096940
89	(2, -9, -6)	297.719828161672	17.254559633954	2.279744203488	1.509882181989	145.105033398321
90	(2, 9, 4)	300.745708053260	17.342021452335	2.302914417531	1.517535639625	146.579810558577
91	(1, 9, 3)	302.100846974693	17.381048500441	2.313291187261	1.520950751097	147.240288833296
92	(3, -1, -9)	302.158285922866	17.382700766074	2.313731017252	1.521095334702	147.268283879997
93	(4, 8, 9)	302.630178343568	17.396269092641	2.317344461534	1.522282648372	147.498278522575
94	(2, -9, -9)	303.345310393612	17.416811143077	2.322820476201	1.524080206617	147.846825210401
95	(5, 4, 9)	305.181711292127	17.469450801102	2.336882436164	1.528686506830	148.741864735840
96	(2, -9, -5)	306.769352516538	17.514832357649	2.349039556840	1.532657677644	149.515661813163
97	(5, 5, 9)	307.654483608226	17.540082200726	2.355817313256	1.534867197270	149.947063972080
98	(1, -9, -3)	309.787850515677	17.600791190048	2.372153245166	1.540179614580	150.986841128570
99	(5, 3, 9)	312.515735600764	17.678114594061	2.393041609401	1.546945897374	152.316379234320
100	(3, -1, 9)	315.824299106262	17.771446173743	2.418376430129	1.555112995936	153.928932959513
101	(3, -8, -9)	319.295794263600	17.868849830462	2.444958875145	1.563636426777	155.620897595721
102	(5, 6, 9)	319.934052549061	17.886700437729	2.449846240678	1.565198466865	155.931976943040
103	(3, 9, 5)	320.688210597366	17.907769559534	2.455621090978	1.567042147161	156.299544429098
104	(2, -9, -4)	321.281219420828	17.924319217779	2.460161965653	1.568490346050	156.588569737440
105	(1, -2, 9)	321.773289217255	17.938040283633	2.463929915114	1.569691025366	156.828398588228
106	(4, 0, 9)	322.099743152572	17.947137464024	2.466429686364	1.570487085704	156.987508276957
107	(0, 2, -9)	324.184787540109	18.005132255557	2.482395595943	1.575561993685	158.003733623313
108	(0, 9, 2)	324.753627361664	18.020921934287	2.486751400170	1.576943689600	158.280979253255
109	(2, 9, 3)	325.881421792573	18.052186066861	2.495387314118	1.579679497277	158.830652580604
110	(4, -4, -9)	326.719437535569	18.075382085466	2.501804291933	1.581709294382	159.239091290000
111	(2, 1, -9)	327.866442696251	18.107082666632	2.510587309116	1.584483294048	159.798127694046
112	(4, -5, -9)	329.164583407961	18.142893468462	2.520527623744	1.587616963800	160.430825732640
113	(5, 2, 9)	329.656556534137	18.156446693506	2.524294832968	1.588802955992	160.670607467520

Rank	Address	q (nm^-2)	sqrt(q)	q/qmin	sqrt ratio	kappa^2
114	(1, 9, 2)	331.408527192029	18.204629279170	2.537710281232	1.593019234420	161.524496717679
115	(4, -3, -9)	334.081088287914	18.277885224717	2.558175010452	1.599429589089	162.827070582080
116	(1, -9, -2)	336.514778590213	18.344339142913	2.576810622981	1.605244723704	164.013221718794
117	(2, -2, 9)	337.777841953316	18.378733415372	2.586482338158	1.608254438252	164.628823483107
118	(2, -9, -3)	341.255428874542	18.473100142492	2.6131114229928	1.616512118708	166.323757171151
119	(4, -6, -9)	341.416525905088	18.477459941915	2.614345005884	1.616893628500	166.402273910000
120	(3, 9, 4)	341.651957858656	18.483829631834	2.616147790182	1.617451016316	166.517020588770
121	(5, 7, 9)	342.020418114631	18.493794043263	2.618969218429	1.618322964809	166.696603648720
122	(4, 9, 9)	344.327067553754	18.556052046536	2.636632034912	1.623770930554	167.820836609517
123	(4, 9, 8)	344.731820688750	18.566955073160	2.639731370348	1.624725013763	168.018108378530
124	(1, 2, -9)	345.012336921878	18.574507716811	2.641879380645	1.625385917450	168.154828588360
125	(3, -9, -7)	349.177780059608	18.686299260678	2.673775626544	1.635168378652	170.185015053774
126	(4, 9, 7)	350.598916373169	18.724286805461	2.684657761245	1.638492527064	170.877659656977
127	(4, -2, -9)	351.249535664995	18.741652426214	2.689639779300	1.640012127791	171.194763608880
128	(3, -9, -8)	352.343316874988	18.770810234910	2.698015242194	1.642563618918	171.727859304879
129	(2, 9, 2)	356.479478081309	18.880664132422	2.729687266167	1.652176523912	173.743774112066
130	(5, 1, 9)	356.604174092246	18.883966058332	2.730642106863	1.652465463138	173.804549435440
131	(0, 9, 1)	358.233274057024	18.927051383061	2.743116691524	1.656235699266	174.598552999993
132	(3, -9, -5)	359.233734077118	18.953462324259	2.750777561631	1.658546822261	175.086165079866
133	(4, 9, 6)	361.928354607011	19.024414698145	2.771411207604	1.664755599962	176.399490444859
134	(4, -7, -9)	363.475265026951	19.065027275799	2.783256438353	1.668309455213	177.153435822080
135	(1, 9, 1)	366.178549958789	19.135792378650	2.803956430664	1.674501845524	178.470984111495
136	(2, -9, -2)	366.691980877678	19.149203139496	2.807887949665	1.675675371206	178.721224114297
137	(4, -1, 9)	368.664023565010	19.200625603480	2.822988565950	1.680175159306	179.682373802495
138	(1, -9, -1)	368.704049214174	19.201667875843	2.823295056257	1.680266364675	179.701881818453
139	(3, -2, 9)	372.198445748292	19.292445302457	2.850052865075	1.688209958825	181.405008307949
140	(3, -9, -4)	372.455224910008	19.299099069905	2.852019112366	1.688792205206	181.530159357064
141	(5, 8, 9)	373.913580304938	19.336845148703	2.863186246509	1.692095223830	182.240944089121
142	(4, -1, -9)	378.224779666811	19.448001945362	2.896198598478	1.701822140671	184.342170370401
143	(4, 9, 5)	378.720135390277	19.460733166823	2.899991709425	1.702936202394	184.583600742175
144	(2, 2, -9)	384.255937362563	19.602447228919	2.942381269219	1.715337071604	187.281683483371
145	(3, 1, -9)	385.523024590994	19.634740247607	2.952083796532	1.718162913269	187.899246430842
146	(1, -3, 9)	387.960371693734	19.696709666686	2.970747410453	1.723585626087	189.087179847740
147	(2, 9, 1)	392.539876919469	19.812619133256	3.005814273677	1.733728431352	191.319175152961
148	(5, 0, 9)	393.358588275092	19.833269732323	3.012083431089	1.735535488283	191.718205138081
149	(6, 4, 9)	394.868885892987	19.871308107243	3.023648304887	1.738864084650	192.454305879638
150	(4, -8, -9)	395.340800773550	19.883178839752	3.027261921152	1.739902848194	192.684311468880
151	(6, 5, 9)	397.344727813942	19.933507664582	3.042606687011	1.744306936009	193.661001204056
152	(2, -9, -1)	397.590875430239	19.939680925989	3.044491524863	1.744847135099	193.780970566877
153	(3, 9, 2)	399.966480029505	19.999161983181	3.062682354974	1.750052100646	194.938811436416
154	(4, 9, 4)	400.974258722967	20.024341655170	3.070399266706	1.752255479862	195.429990548925
155	(2, -3, 9)	403.961854824938	20.098802323147	3.093276328466	1.758771255299	196.886108654441
156	(1, -9, 0)	406.355662387557	20.158265361572	3.111606544995	1.763974644091	198.052821427546
157	(1, 9, 0)	406.410915274972	20.159635792220	3.112029635558	1.764094565367	198.079751014747
158	(1, 3, -9)	411.205558608070	20.278204028170	3.148743886046	1.774470029627	200.416602024228
159	(4, 0, -9)	415.006820293364	20.371716184292	3.177851467985	1.782652929761	202.269290866641
160	(3, -9, -2)	415.285234224059	20.378548383633	3.179983380221	1.783250790052	202.404986439763
161	(5, 9, 9)	415.613539119981	20.386601951281	3.182497324919	1.783955527730	202.564998264241
162	(5, 9, 8)	417.308668326376	20.428134235078	3.195477518433	1.787589863037	203.391183680333
163	(6, 2, 9)	419.337591925284	20.477734052509	3.211013691628	1.791930158133	204.380056434963
164	(5, -4, -9)	421.192773523861	20.522981594395	3.225219461939	1.795889601824	205.284249445833
165	(4, -9, -7)	422.638964821902	20.558184862042	3.236293451364	1.798970108524	205.989105544597
166	(5, -5, -9)	423.634849791395	20.582391741277	3.243919288718	1.801088362274	206.474487800296
167	(4, -9, -6)	423.645394484546	20.582647897794	3.244000033098	1.801110777575	206.479627155847
168	(5, 9, 7)	424.466140082195	20.602576054518	3.250284767408	1.802854616271	206.879648605859
169	(4, -2, 9)	425.035100602183	20.616379425161	3.254641495865	1.804062497771	207.156953062754
170	(4, -9, -8)	427.094877708682	20.666273919328	3.270413925091	1.808428578930	208.160863442782
171	(5, -3, -9)	428.557493881062	20.701630222788	3.281613685489	1.811522477224	208.873724826091

Rank	Address	q (nm^-2)	sqrt(q)	q/qmin	sqrt ratio	kappa^2
172	(4, 9, 3)	428.690724605079	20.704847852739	3.282633879450	1.811804040025	208.938659865110
173	(4, -9, -5)	430.114166696613	20.739193974131	3.293533670294	1.814809541052	209.632428276532
174	(6, 7, 9)	431.716801530060	20.777795877572	3.305805602246	1.818187449700	210.413533057052
175	(2, -9, 0)	433.952112532222	20.831517288288	3.322922155522	1.822888410058	211.502996528891
176	(2, 9, 0)	434.062618307052	20.834169489256	3.323768336649	1.823120494276	211.556855703291
177	(5, -6, -9)	435.883722683666	20.877828495408	3.337713165827	1.826940931127	212.444439889478
178	(4, -9, -9)	437.013133144886	20.904859079766	3.346361454280	1.829306276784	212.994900850401
179	(5, 9, 6)	437.085954387438	20.906600737266	3.346919071845	1.829458682738	213.030393040820
180	(3, 9, 1)	437.317254939064	20.912131764578	3.348690220563	1.829942682316	213.143126124390
181	(5, -1, 9)	439.919799082673	20.974265161923	3.368618805643	1.835379744261	214.411574575442
182	(0, 9, -1)	441.579595096014	21.013795352007	3.381328440617	1.838838883811	215.220539021771
183	(3, 2, -9)	441.91558862162	21.021788431581	3.383901261667	1.839538328404	215.384298308345
184	(4, -9, -4)	442.045281458103	21.024872923709	3.384894362951	1.839808240810	215.447508906651
185	(3, -9, -1)	444.893752705219	21.092504656992	3.406706097341	1.845726441632	216.835819245264
186	(5, -2, -9)	445.729010862999	21.112295253311	3.413101959368	1.847458242929	217.242913941070
187	(6, 1, 9)	446.282139878537	21.125390881083	3.417337460493	1.848604192490	217.512502314705
188	(1, -9, 1)	449.469618110364	21.200698528831	3.441745089194	1.855194083969	219.066040546074
189	(2, 3, -9)	450.452228653610	21.223859890548	3.449269279652	1.857220848379	219.544953007416
190	(1, 9, -1)	452.105623140578	21.262775527682	3.461929895913	1.860626210692	220.350797427432
191	(5, 9, 5)	455.168111242103	21.334669232076	3.485380431744	1.866917360716	221.843416985215
192	(5, -7, -9)	457.939392200673	21.399518503945	3.506601093265	1.872592078715	223.194105713380
193	(4, -9, -3)	459.438738769017	21.434522125977	3.518082111070	1.875655115172	223.924869046204
194	(4, 1, -9)	461.595657544652	21.484777344544	3.534598387821	1.880052761978	224.976125097602
195	(4, 9, 2)	461.869533036615	21.491150109676	3.536695547655	1.880610418895	225.109608690729
196	(6, 8, 9)	463.613033325223	21.531675116563	3.550046135358	1.884156611155	225.959369585630
197	(1, -4, 9)	463.954250794948	21.539597275598	3.552658956121	1.884849849755	226.125674841972
198	(0, 4, -9)	466.371888327515	21.595645124134	3.571171647012	1.889754388012	227.304002053412
199	(5, -1, -9)	472.707324469672	21.741833512141	3.619684283577	1.902546788801	230.391816790768
200	(2, -9, 1)	475.775692183629	21.812283057572	3.643179841643	1.908711565859	231.887302000340
815	(5, 9, -9)	1281.864274896880	35.803132193942	9.815680293781	3.132998610562	624.765100697356
816	(7, -9, 8)	1284.094314878390	35.834261745966	9.832756485020	3.135722641596	625.851995137658
817	(7, 7, -9)	1285.840868273719	35.858623346048	9.846130451423	3.137854434390	626.703244079746
818	(9, 8, -4)	1298.212087795704	36.030710342647	9.940861179199	3.152913125857	632.732826432389
819	(6, 9, -8)	1298.948238189226	36.040924491323	9.946498138628	3.153806927925	633.091617205909
820	(9, -9, 4)	1306.714537097473	36.148506706328	10.005967388721	3.163221046453	636.876817101465
821	(9, -8, 6)	1310.345262465954	36.198691446874	10.033769114808	3.167612525990	638.646388611386
822	(8, 9, -5)	1314.970036058012	36.262515578184	10.069182613647	3.173197537760	640.900447169323
823	(9, 4, -9)	1316.852923771796	36.288468192689	10.083600539312	3.175468554294	641.818143804717
824	(9, 5, -8)	1321.489243389449	36.352293509343	10.119102450082	3.181053669790	644.077829755494
825	(9, -7, 8)	1327.110749981533	36.429531289622	10.162148279939	3.187812459970	646.817683888925
826	(8, 6, -9)	1330.154227673046	36.471279490485	10.185453246453	3.191465689374	648.301038003432
827	(9, 6, -7)	1332.133999843596	36.498410922170	10.200613050077	3.193839859805	649.265954948006
828	(7, 9, -7)	1337.329842947111	36.569520682491	10.240399426653	3.200062409806	651.798345859701
829	(8, -9, 7)	1339.546341736375	36.599813411223	10.257371928277	3.202713213555	652.878640487135
830	(9, 7, -6)	1348.787193134236	36.725838222350	10.328132339297	3.213741174908	657.382519382253
831	(9, -9, 5)	1361.354854446774	36.896542581206	10.424367290142	3.228678876900	663.507845081098
832	(7, -9, 9)	1363.164754568184	36.921061124623	10.438226324443	3.230824403220	664.389968449188
833	(8, -8, 9)	1364.637751432842	36.941003660334	10.449505573409	3.232569500167	665.107889256013
834	(9, -8, 7)	1371.279913745269	37.030796828387	10.500366918869	3.240426965520	668.345198608679
835	(9, 8, -5)	1371.448823261370	37.033077420886	10.501660317742	3.240626531667	668.427523058234
836	(9, -7, 9)	1394.339735190863	37.340858790216	10.676943986640	3.267559331770	679.584275903877
837	(8, 9, -6)	1397.009089170535	37.376584771358	10.697384157857	3.270685579180	680.885286658733
838	(9, 9, -4)	1400.118890224998	37.418162571471	10.721196985412	3.274323897450	682.400965975950
839	(7, 8, -9)	1401.086490712728	37.431089894802	10.728606238658	3.275455119317	682.872562718281
840	(9, 5, -9)	1402.684295546311	37.452427098204	10.740841185622	3.277322258433	683.651313415447
841	(8, -9, 8)	1411.864062805347	37.574779610869	10.811133854160	3.288028870640	688.125420642151
842	(9, 6, -8)	1412.497060619866	37.583201840980	10.815980938439	3.288765868596	688.433936099734
843	(9, 7, -7)	1428.318262529917	37.793098080601	10.937129380480	3.307133105951	696.144998025757

Rank	Address	q (nm ⁻²)	sqrt(q)	q/qmin	sqrt ratio	kappa ²
844	(7, 9, -8)	1429.003205087081	37.802158735806	10.942374223708	3.307925970107	696.478830720901
845	(8, 7, -9)	1435.596123092176	37.889261316264	10.992858488390	3.315547992171	699.692138995425
846	(9, -8, 8)	1437.676907574008	37.916710136482	11.008791778391	3.317949936089	700.706288115406
847	(9, 8, -6)	1450.147901276460	38.080807518702	11.104286511747	3.332309486189	706.784499193513
848	(9, 9, -5)	1477.985976859498	38.444583192688	11.317452332238	3.364142139125	720.352439603005
849	(8, 9, -7)	1484.510484832481	38.529345761802	11.367412757528	3.371559395521	723.532405657578
850	(9, -9, 7)	1487.022516793647	38.561930926675	11.386648259368	3.374410801809	724.756739568669
851	(8, -9, 9)	1489.644126423741	38.595908156484	11.406722835505	3.377384022510	726.034480306601
852	(9, -8, 9)	1509.536243952170	38.852750790030	11.559043693375	3.399859363764	735.729657131567
853	(9, 7, -8)	1513.311674475020	38.901306847907	11.587953477125	3.404108323354	737.569756178694
854	(7, 9, -9)	1526.138909776474	39.065827903380	11.686176076224	3.418504947521	743.821595091536
855	(9, 8, -7)	1534.309321840973	39.170260681300	11.748739761213	3.427643470551	747.803754838227
856	(8, 8, -9)	1550.844815136041	39.380767071453	11.875357780658	3.446064099906	755.862953722138
857	(9, -9, 8)	1558.049861791219	39.472140324427	11.930529327173	3.454059832599	759.374606076606
858	(8, 9, -8)	1577.474223043851	39.717429713463	12.079268412660	3.475524192501	768.841804165857
859	(9, 7, -9)	1603.767428969547	40.047065172988	12.280604629231	3.504369362557	781.656793841067
860	(9, 8, -8)	1623.933084954910	40.298053116186	12.435020066140	3.526332381688	791.485289992375
861	(9, 9, -7)	1650.107177776766	40.621511268991	12.635444192275	3.554636998665	804.242225385418
862	(8, 9, -9)	1675.900303804643	40.937761343345	12.832951123254	3.582310863570	816.813482183570
863	(9, 8, -9)	1719.019190618270	41.461056313344	13.163127426530	3.628102455352	837.829104655958
864	(9, 9, -8)	1744.361292059535	41.765551499526	13.357180705485	3.654747693820	850.180537540776

Appendix E. Symbols and notation

Symbol	Meaning
B	Locked Cell3 basis matrix; columns are primitive directions a,b,c.
M	Direct Gram metric $B^T B$.
R	Reciprocal metric M^{-1} .
V	Cell3 volume $ \det B $.
C_k	Oriented k-cell cochain space.
B_k	Cellular boundary matrix lowering cell dimension.
d_k	Coboundary $B_{(k+1)}^T$.
H_k	Metric Hodge-star matrix in form degree k.
δ_k	Metric adjoint of $d_{(k-1)}$.
Δ_k	k-form Hodge Laplacian.
ξ_j	One-dimensional incidence symbol $2 \sin(\pi j/20)$.
$\Lambda(j)$	Open-complex modal Hodge eigenvalue $\xi^T R \xi$.
s	Primitive integer outer-shell address.
q(s)	Reciprocal quadratic shell value $s^T R s$.
κ_s	Periodic 9B reciprocal wavenumber magnitude.

Source files

File	SHA-256
K-Field_Lucas_Supergroup_metric_Hodge_program_1785236325.json	3819b8623aea28431b13e4f74fbb7bc41db761d88773617c6b42049d236f8e70
K-Field_Lucas_Supergroup_Analysis_1785208014.json	37efaa688f9cb496ae16c76af6f7889bd78e10e4e27c15cee9b3e2cd3331f6b5
K-Field_Mass-Bridge_is_now_a_locked_projective_ratio_system_1785176592(1).json	a5ba3ddf21a20e4528c19609a3061fceca60f53f76224f101c9779c24809d7ea
K-Field_mass_bridge_drives_volume_filling_hierarchy_1785204375.json	811c98f49447534c27d14fd77c779d756e4acc73bf44bfbf93ae7bebf81e79b2
K-Field_Lucas_Supergroup_metric_Hodge_operators_1785236325.zip	bbecb20c707dec7c8f4b15988ff2e6e16b27a2fce0ff2ec4a12be264dee299f3

PDF filename: K-Field_Lucas_Supergroup_exact_deterministic_reciprocal_field_spectrum_1785237526.pdf

Machine-readable filename: K-Field_Lucas_Supergroup_exact_deterministic_reciprocal_field_spectrum_1785237526.json